

ME5412 Robotics for Healthcare

Principles for Assistive Technologies

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Principles for Assistive Technologies

- WHO definitions
- Definition of Assistive Technology
- HAAT model
- Principles of Universal Design and user-centred design
- Assistive Technology characterization
- Robotics as Assistive Technology

Disability key facts

- An estimated **1.3 billion** people experience significant disability. This represents **16%** of the world's population, or 1 in 6 of us.
- Some persons with disabilities die up to 20 years earlier than those without disabilities.
- Persons with disabilities have twice the risk of developing conditions such as depression, asthma, diabetes, stroke, obesity or poor oral health.
- Persons with disabilities face many health inequities.
- Persons with disabilities find **inaccessible** and unaffordable transportation 15 times more difficult than for those without disabilities.
- Health inequities arise from unfair conditions faced by persons with disabilities, including **stigma, discrimination, poverty, exclusion from education and employment**, and **barriers** faced in the health system itself.

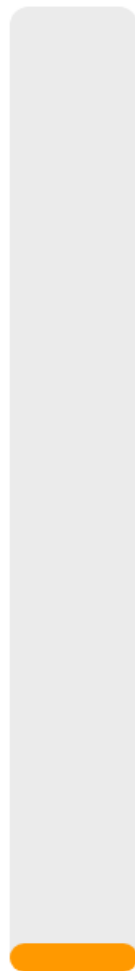


<https://wecapable.com/types-of-disabilities-list/>

Is disability increasing or decreasing?

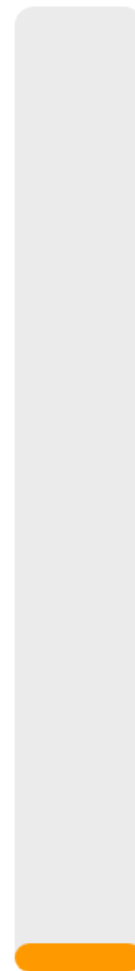


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Increasing

0%



Decreasing

Possible reasons for disability increase?



Nobody has responded yet.

Hang tight! Responses are coming in.

Increasing disability

Disability is increasing due to:

- longer life expectation, with increased prevalence of health conditions and diseases
- increased risks of traumas, in car accidents and hazardous sports
- increased survival rates from traumatic injury

Increased need for personal assistance

Disability and ageing

Fig. 2.2. Age-specific disability prevalence, derived from multidomain functioning levels in 59 countries, by country income level and sex

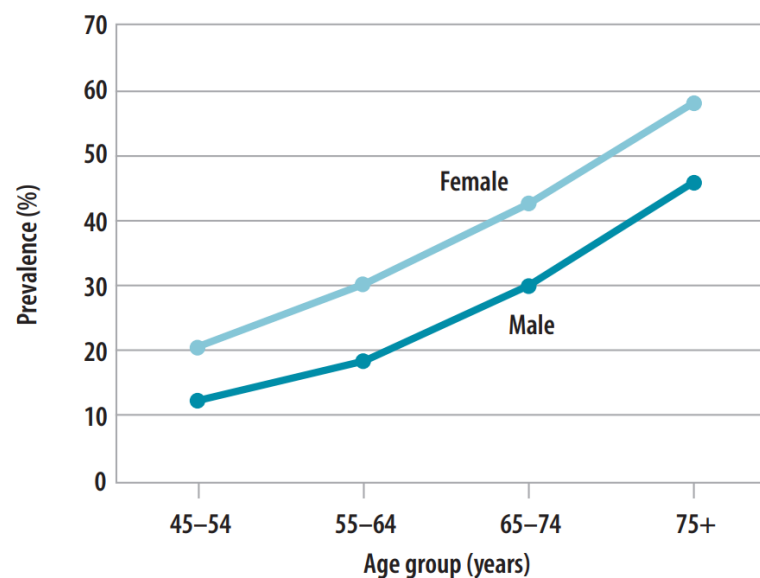
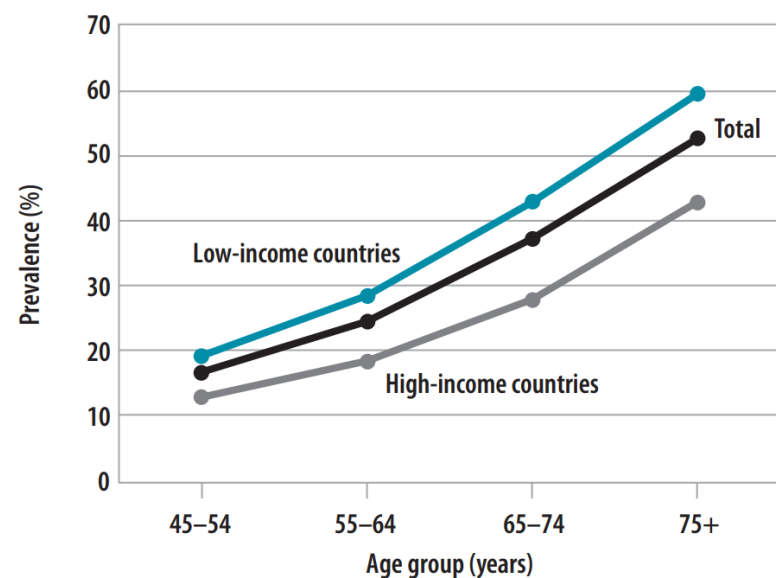
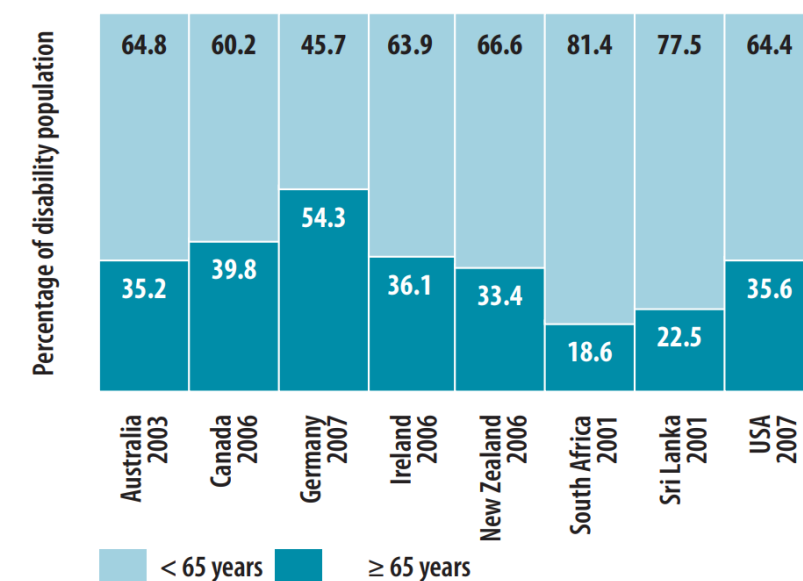


Fig. 2.3. Distribution of ages within disability populations



Can you define disability?



Nobody has responded yet.

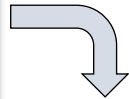
Hang tight! Responses are coming in.

WHO international classifications

Objectives: to provide a uniform language for diagnostic practice, epidemiological studies, clinical assessment and rehabilitation programs

ICIDH

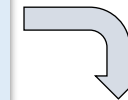
INTERNATIONAL CLASSIFICATION OF
IMPAIRMENTS, DISABILITY AND HANDICAP
1980



ICIDH-2

INTERNATIONAL CLASSIFICATION OF
IMPAIRMENTS, OF PERSONAL ACTIVITIES
AND OF SOCIAL PARTICIPATION

1997



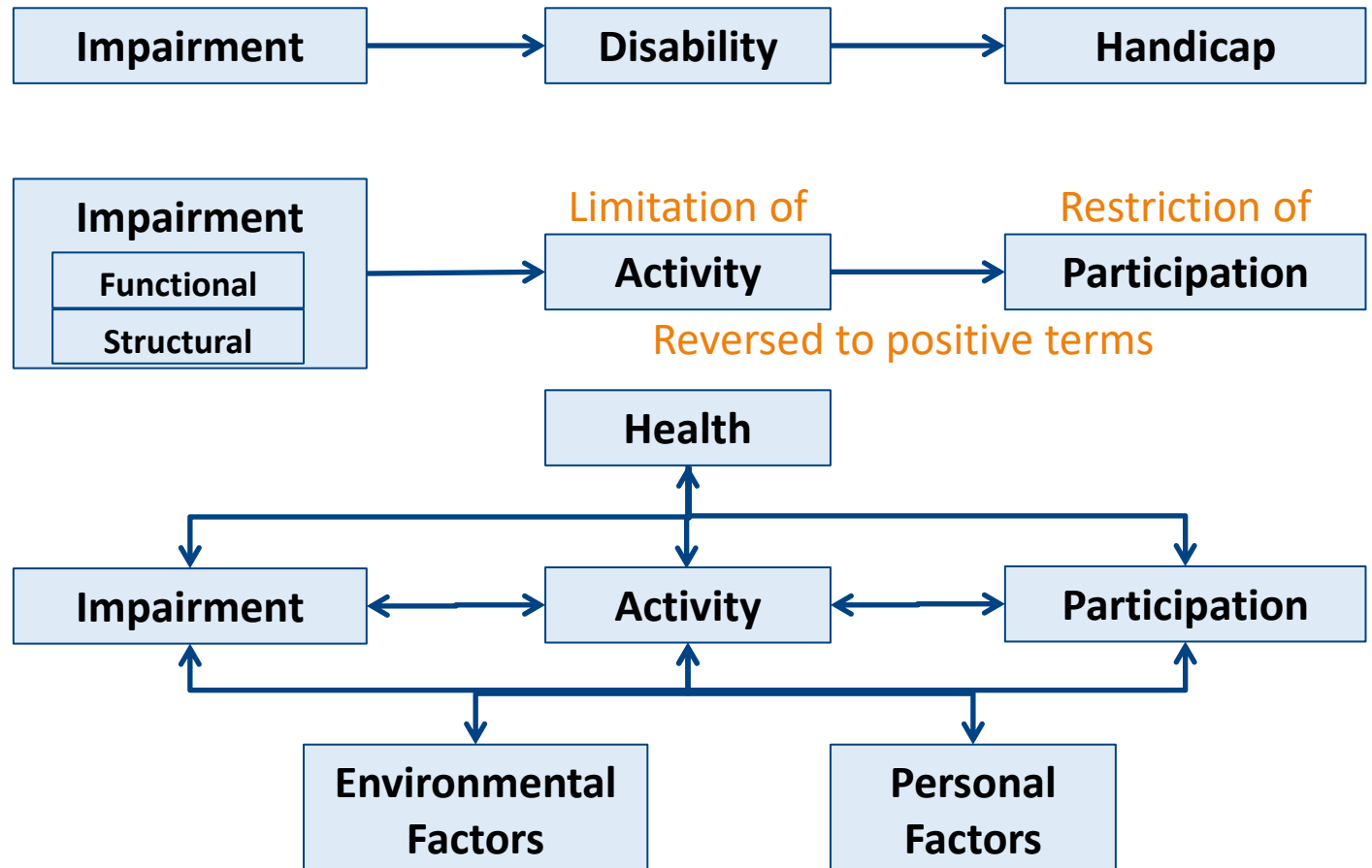
ICF

INTERNATIONAL CLASSIFICATION
OF FUNCTIONING, DISABILITY
AND HEALTH

2001

WHO international classifications

- ICIDH (1980)
- ICIDH2 (1997)
- IFC (2001)



WHO International Classification of Impairments, Disability and Handicap

ICIDH (1980)



- A disease leads to an **impairment**: abnormality of a body structure or appearance or malfunction of an organ or system)
 - An impairment leads to a **disability**: limitation to functional performance and activity of the individual
 - A disability leads to a **handicap**: disadvantage experienced by the individual as a result of the individual's disability and impairments
- Focus on the impairment that prevents an individual from performing



Do they have

- Impairment?
- Disability?
- Handicap?



WHO International Classification of Impairments, Personal Activities and Social Participation

ICIDH2 (1997)

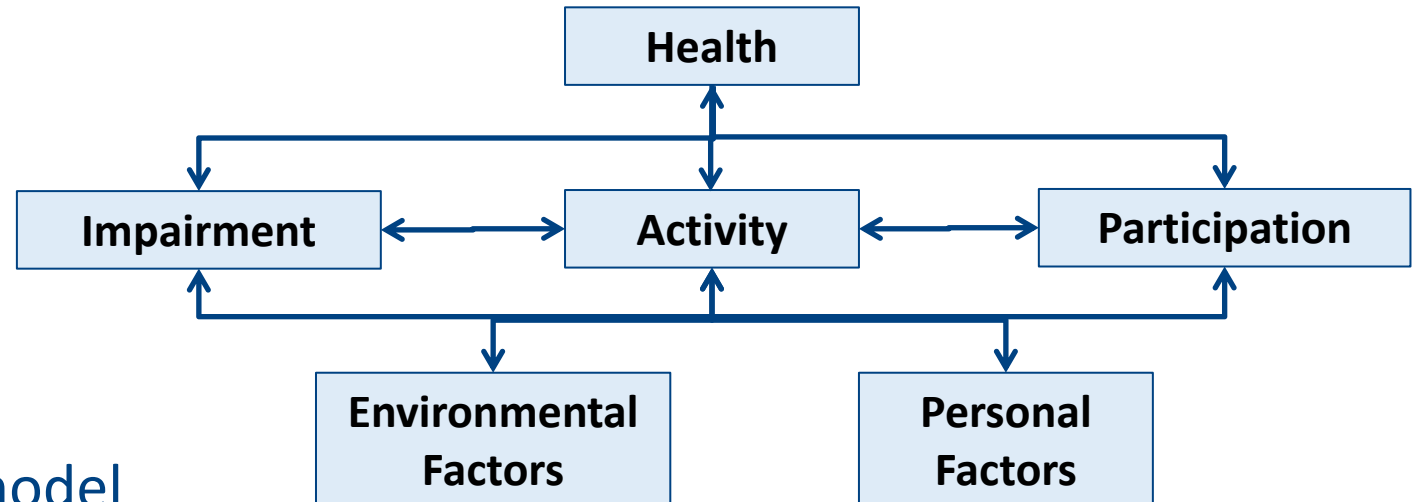


- Impairment is considered in its functional and structural aspects
- Disability and handicap are replaced by positive terms, and expressed as limitations:
 - Activity limitation
 - Participation restriction

Focus on the barriers erected by society that hinder full participation of people with disabilities

WHO International Classification of Functioning, Disability and Health

ICF (2001)



Focus on a bio-psycho-social model

- **Disability** covers impairment, activity and participation
- Disability reflects the negative aspects of the interaction between the individual's health condition and contextual (environmental and personal) factors
- **Functioning** covers body functions, body structures, activities, participation
- Functioning denoted positive or neutral aspects of the interaction between the individual's health condition and contextual (environmental and personal) factors

ICF in the WHO World Report on Disability

Box 1.1. New emphasis on environmental factors

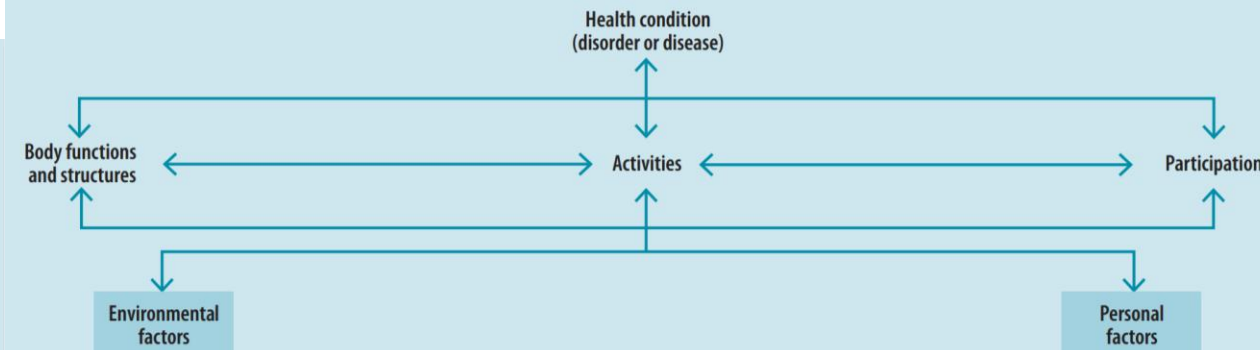
The *International Classification of Functioning, Disability and Health* (ICF) (17) advanced the understanding and measurement of disability. It was developed through a long process involving academics, clinicians, and – importantly – persons with disabilities (18). The ICF emphasizes environmental factors in creating disability, which is the main difference between this new classification and the previous *International Classification of Impairments, Disabilities, and Handicaps* (ICIDH). In the ICF, problems with human functioning are categorized in three interconnected areas:

- **impairments** are problems in body function or alterations in body structure – for example, paralysis or blindness;
- **activity limitations** are difficulties in executing activities – for example, walking or eating;
- **participation restrictions** are problems with involvement in any area of life – for example, facing discrimination in employment or transportation.

Disability refers to difficulties encountered in any or all three areas of functioning. The ICF can also be used to understand and measure the positive aspects of functioning such as body functions, activities, participation and environmental facilitation. The ICF adopts neutral language and does not distinguish between the type and cause of disability – for instance, between “physical” and “mental” health. “**Health conditions**” are diseases, injuries, and disorders, while “**impairments**” are specific decrements in body functions and structures, often identified as symptoms or signs of health conditions.

Disability arises from the interaction of health conditions with contextual factors – environmental and personal factors as shown in the figure below.

Representation of the International Classification of Functioning, Disability and Health



The ICF contains a classification of **environmental factors** describing the world in which people with different levels of functioning must live and act. These factors can be either facilitators or barriers. Environmental factors include: products and technology; the natural and built environment; support and relationships; attitudes; and services, systems, and policies.

The ICF also recognizes **personal factors**, such as motivation and self-esteem, which can influence how much a person participates in society. However, these factors are not yet conceptualized or classified. It further distinguishes between a person’s **capacities** to perform actions and the actual **performance** of those actions in real life, a subtle difference that helps illuminate the effect of environment and how performance might be improved by modifying the environment.

The ICF is universal because it covers all human functioning and treats disability as a continuum rather than categorizing people with disabilities as a separate group: disability is a matter of more or less, not yes or no. However, policy-making and service delivery might require thresholds to be set for impairment severity, activity limitations, or participation restriction.

It is useful for a range of purposes – research, surveillance, and reporting – related to describing and measuring health and disability, including: assessing individual functioning, goal setting, treatment, and monitoring; measuring outcomes and evaluating services; determining eligibility for welfare benefits; and developing health and disability surveys.

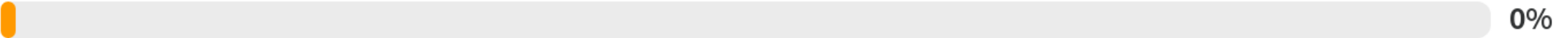
Which are true about disability?



Disability is an impairment of body functions or structures



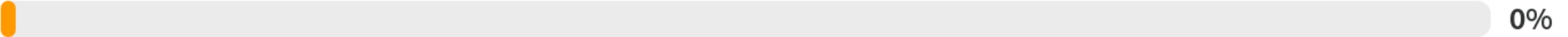
Disability concerns impairment, activity limitations and participation restrictions



Disability is a health condition (disorder or disease)



Disability restricts a person's participation in social life



Disability and handicap are synonymous



Disability limits a person's activities in daily life



Common types of assistance and support

- **Community support and independent living** – assistance with self-care, household care, mobility, leisure, and community participation;
- **residential support services** – independent housing and congregate living in group homes and institutional settings;
- **respite services** – short-term breaks for caregivers and people with disabilities;
- **support in education or employment** – such as a classroom assistant for a child with a disability, or personal support in the workplace;
- **communication support** – such as sign-language interpreters;
- **community access** – including day care centres;
- **information and advice services** – including professional, peer support, advocacy, and supported decision-making;
- **assistance animals** – such as dogs trained to guide people with a visual impairment.



UN Convention on the rights of persons with disabilities (CRPD)

Article 26, Habilitation and Rehabilitation:

“... appropriate measures, including through peer support, to enable persons with disabilities to attain and maintain their maximum independence, full physical, mental, social and vocational ability, and full inclusion and participation in all aspects of life”.

- Services should be provided in the community, not in segregated settings
- Person-centred services are preferable, so that individuals are involved in decisions about the support they receive and have maximum control over their lives.

A few definitions around rehabilitation

- **Rehabilitation** addresses the impact of a health condition on a person's everyday life by optimizing their functioning and reducing their experience of disability (WHO 2023)
- **Rehabilitation**: process aimed at enabling people with disability to reach and maintain their optimal physical, sensory, intellectual, psychological and social functional levels (WHO 2016)
- **Rehabilitation**: a set of measures that assist individuals who experience, or are likely to experience, disability to achieve and maintain optimal functioning in interaction with their environments (WHO Report on Disability, 2011)

A few definitions around rehabilitation

- **Rehabilitation** aims at filling the gap between the capabilities of an individual with disabilities and the requirements of an activity the person wants to perform, through one or a combination of the following four strategies (EUSTAT 1999):
 - Receive **training** to improve the individual's capabilities
 - Change the **environment** to reduce the requirements of the activities
 - Use adequate **assistive technology** to perform the activity
 - Have a personal **helper** do or help with the activity
- **Rehabilitation** is the (re)integration of an individual with a disability into society (Robinson, 1993)

A few definitions around rehabilitation

- **Rehabilitation Therapy** is concerned with restoring and compensating for the loss of functioning, and preventing or slowing deterioration in functioning in every area of a person's life (WHO Report on Disability, 2011).
- **Rehabilitation Engineering** is the application of science and technology to ameliorate the handicaps of individuals with disabilities (Reswick, 1982)
- **Rehabilitation Technology** (or **Assistive Technology**) is the selection, design, or manufacture of augmentative or assistive devices that are appropriate for the individual with a disability
- **Assistive technology** enables and promotes inclusion and participation, especially of persons with disability, aging populations, and people with non-communicable diseases. The primary purpose of assistive products is to maintain or improve an individual's functioning and independence, thereby promoting their well-being. They enable people to live healthy, productive, independent and dignified lives, and to participate in education, the labour market and civic life (WHO Report on Disability, 2011).

Assistive Technology (AT)

Technology* designed to be utilized in an assistive technology device or assistive technology service

- **Assistive technology device:** any item, piece of equipment, or product system, whether acquired commercially, modified, or customized, that is used to increase, maintain, or improve functional capabilities of individuals with disabilities.

United States Assistive Technology Act (1998, amended 2004)

- **Assistive technology device:** any product, instrument, equipment, or technology adapted or specially designed for improving the functioning of a disabled person.

WHO (2001)

Common
meaning

Our
focus

- **Assistive technology service:** any service that directly assists an individual with disabilities in the selection, acquisition, or use of an assistive technology device.

United States Assistive Technology Act (1998, amended 2004)

*Technology: the practical application of knowledge especially in a particular area – not a machine or piece of equipment that is created by technology → technology device

Can you name an assistive technology device?

 0

Nobody has responded yet.

Hang tight! Responses are coming in.

Examples of assistive devices

- Crutches, prostheses, orthoses, wheelchairs, and tricycles for people with mobility impairments;
- hearing aids and cochlear implants for those with hearing impairments;
- white canes, magnifiers, ocular devices, talking books, and software for screen magnification and reading for people with visual impairments;
- communication boards and speech synthesizers for people with speech impairments;
- devices such as day calendars with symbol pictures for people with cognitive impairment.

Assistive technologies, when appropriate to the user and the user's environment, have been shown to be powerful tools to increase independence and improve participation.

- Should respond to user's needs
- Do not improve impairment
- Maximise activity and participation



Rehabilitation process

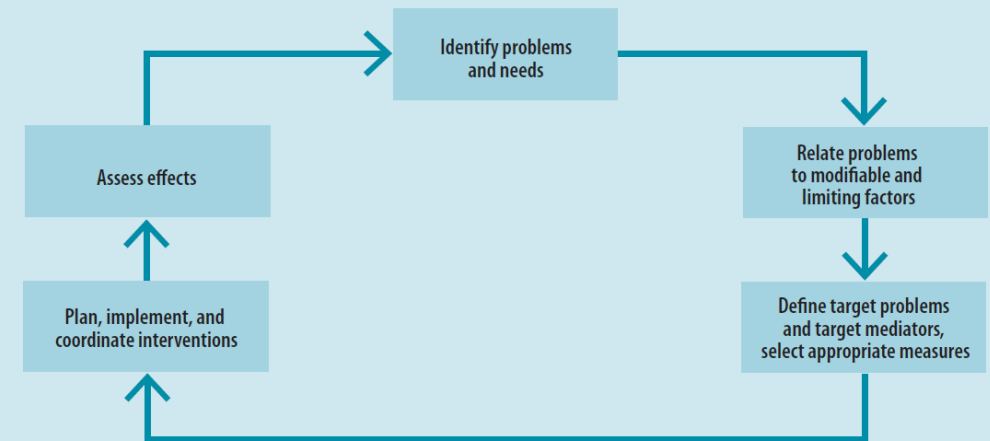
Box 4.1. What is rehabilitation?

This Report defines **rehabilitation** as “a set of measures that assist individuals who experience, or are likely to experience, disability to achieve and maintain optimal functioning in interaction with their environments”. A distinction is sometimes made between habilitation, which aims to help those who acquire disabilities congenitally or early in life to develop maximal functioning; and rehabilitation, where those who have experienced a loss in function are assisted to regain maximal functioning (2). In this chapter the term “rehabilitation” covers both types of intervention. Although the concept of rehabilitation is broad, not everything to do with disability can be included in the term. Rehabilitation targets improvements in individual functioning – say, by improving a person’s ability to eat and drink independently. Rehabilitation also includes making changes to the individual’s environment – for example, by installing a toilet handrail. But barrier removal initiatives at societal level, such as fitting a ramp to a public building, are not considered rehabilitation in this Report.

Rehabilitation reduces the impact of a broad range of health conditions. Typically rehabilitation occurs for a specific period of time, but can involve single or multiple interventions delivered by an individual or a team of rehabilitation workers, and can be needed from the acute or initial phase immediately following recognition of a health condition through to post-acute and maintenance phases.

Rehabilitation involves identification of a person’s problems and needs, relating the problems to relevant factors of the person and the environment, defining rehabilitation goals, planning and implementing the measures, and assessing the effects (see figure below). Educating people with disabilities is essential for developing knowledge and skills for self-help, care, management, and decision-making. People with disabilities and their families experience better health and functioning when they are partners in rehabilitation (3–9).

The rehabilitation process



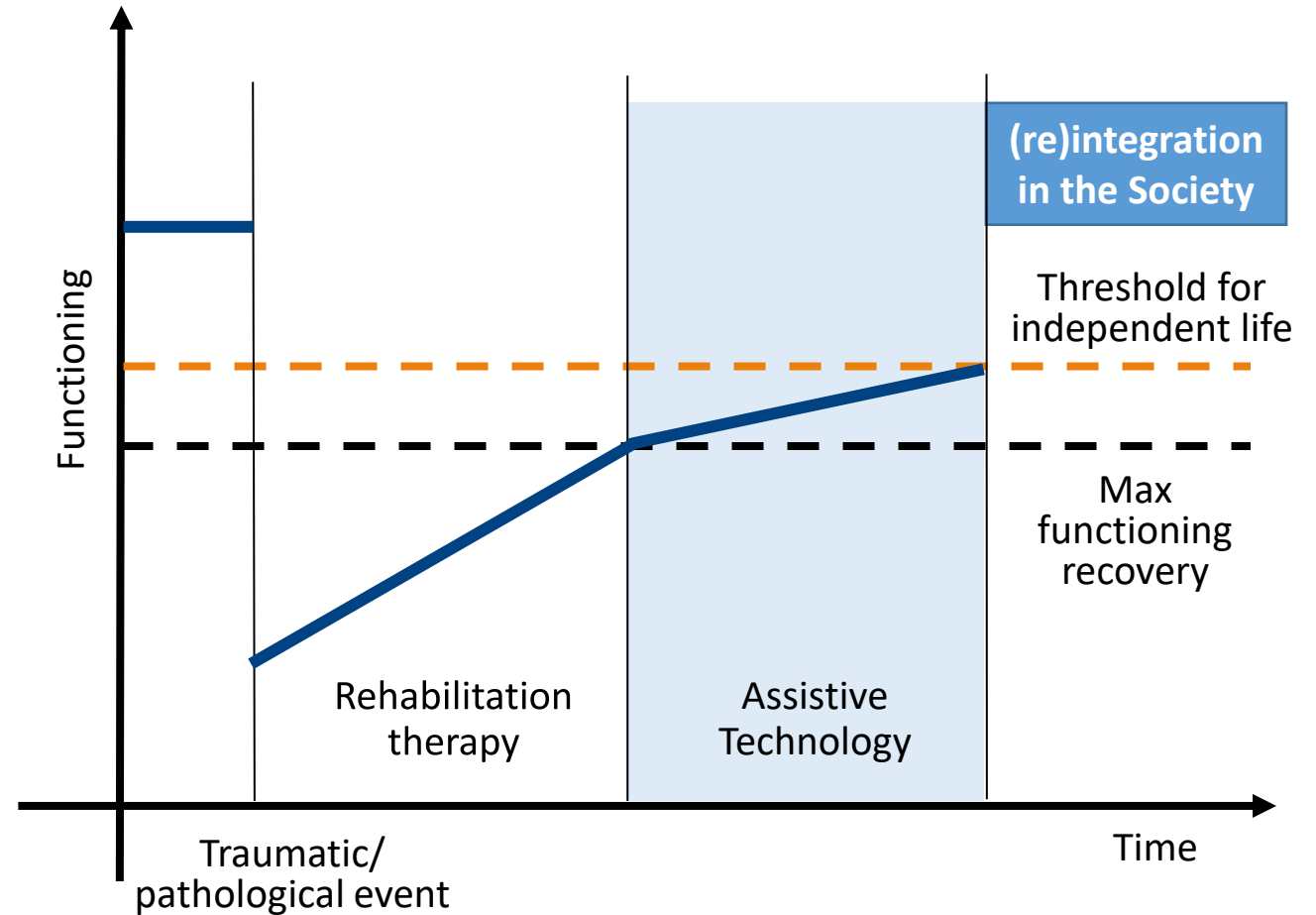
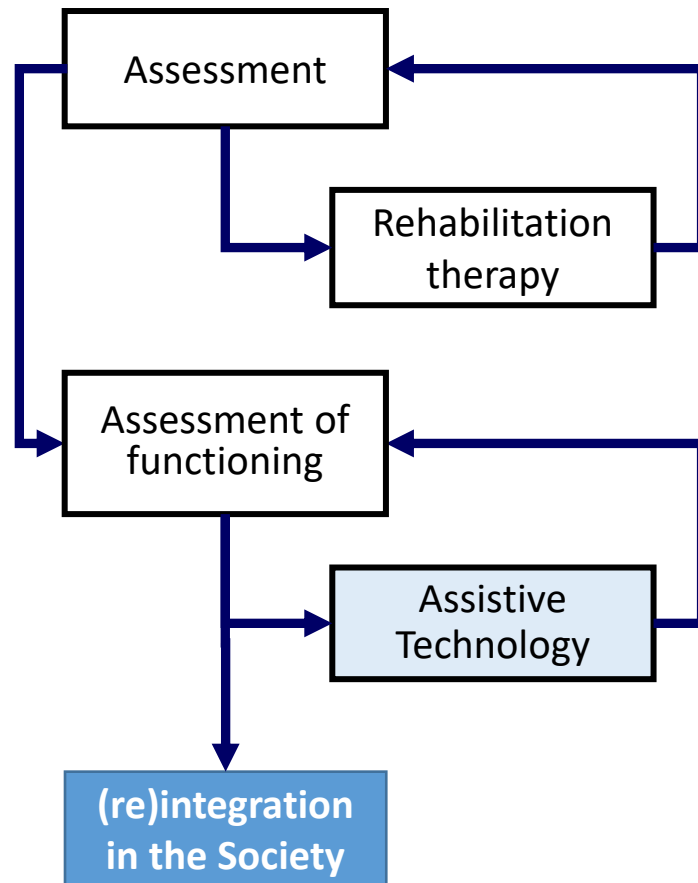
Source: A modified version of the Rehabilitation Cycle from (10).

Rehabilitation – provided along a continuum of care ranging from hospital care to rehabilitation in the community (12) – can improve health outcomes, reduce costs by shortening hospital stays (15–17), reduce disability, and improve quality of life (18–21). Rehabilitation need not be expensive.

Rehabilitation is cross-sectoral and may be carried out by health professionals in conjunction with specialists in education, employment, social welfare, and other fields. In resource-poor contexts it may involve non-specialist workers – for example, community-based rehabilitation workers in addition to family, friends, and community groups.

Rehabilitation that begins early produces better functional outcomes for almost all health conditions associated with disability (18–30). The effectiveness of early intervention is particularly marked for children with, or at risk of, developmental delays (27, 28, 31, 32), and has been proven to increase educational and developmental gains (4, 27).

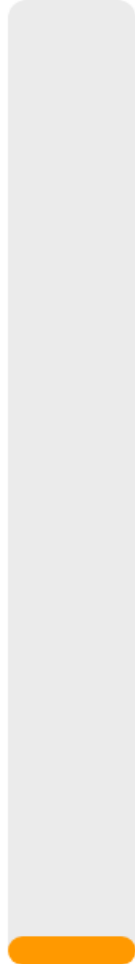
Rehabilitation process



Does assistive technology improve impairment?

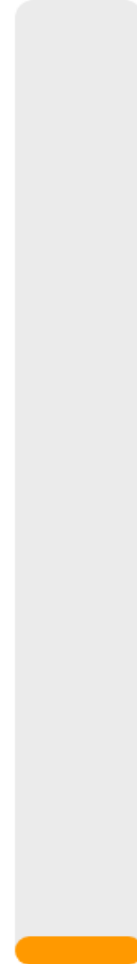
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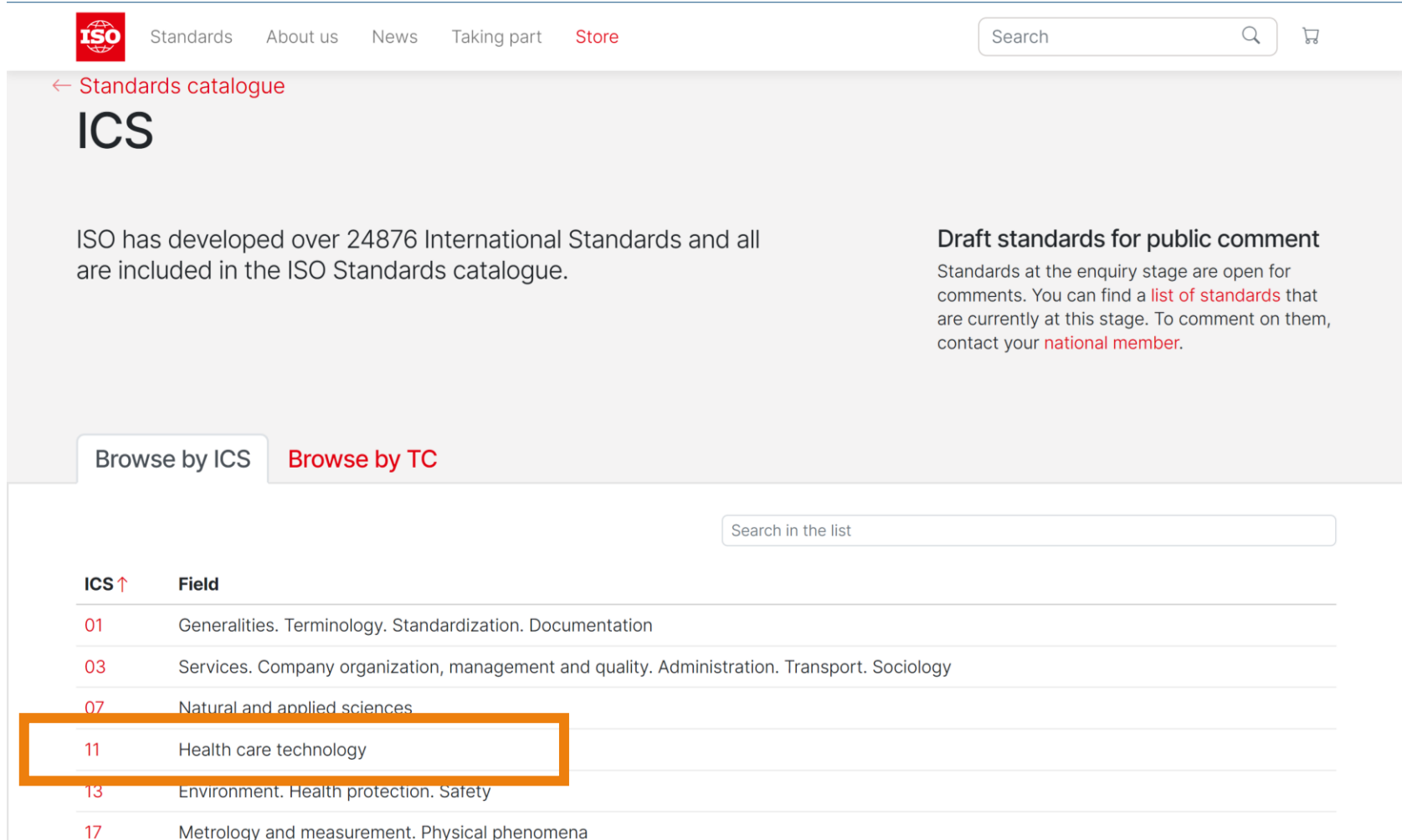
Yes

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No

ISO Classification of Assistive Technology

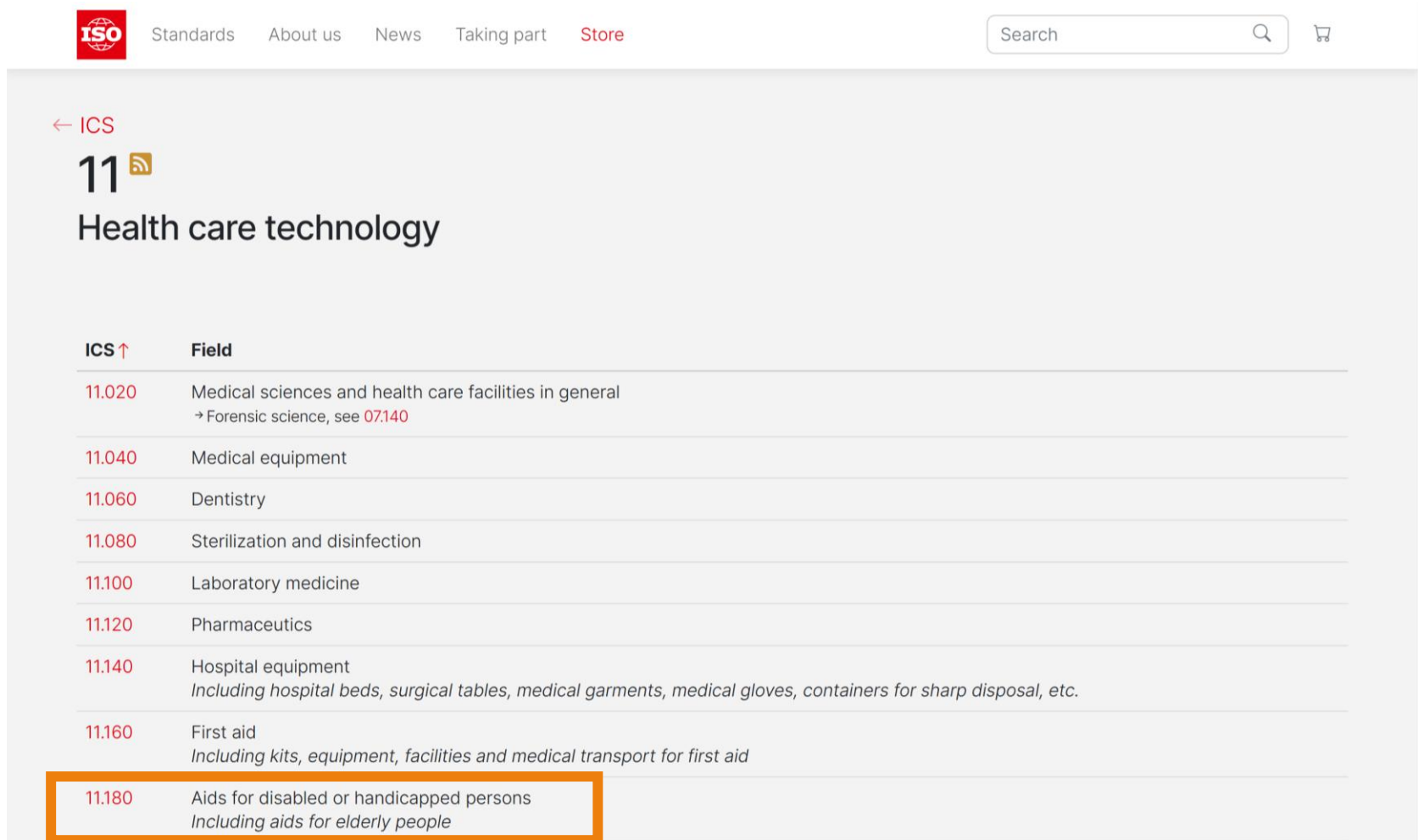


The screenshot shows the ISO Standards catalogue website. At the top, there is a navigation bar with the ISO logo and links for Standards, About us, News, Taking part, and Store. A search bar is also present. Below the navigation bar, the page title is "← Standards catalogue" followed by "ICS". A paragraph states: "ISO has developed over 24876 International Standards and all are included in the ISO Standards catalogue." To the right, there is a section titled "Draft standards for public comment" with a description. Below this, there are two tabs: "Browse by ICS" (selected) and "Browse by TC". A search bar labeled "Search in the list" is positioned above a table. The table has two columns: "ICS ↑" and "Field". The table lists several ICS categories, with "11 Health care technology" highlighted by an orange box.

ICS ↑	Field
01	Generalities. Terminology. Standardization. Documentation
03	Services. Company organization, management and quality. Administration. Transport. Sociology
07	Natural and applied sciences
11	Health care technology
13	Environment. Health protection. Safety
17	Metrology and measurement. Physical phenomena

ISO 9999:2022

ISO Classification of Assistive Technology



← ICS

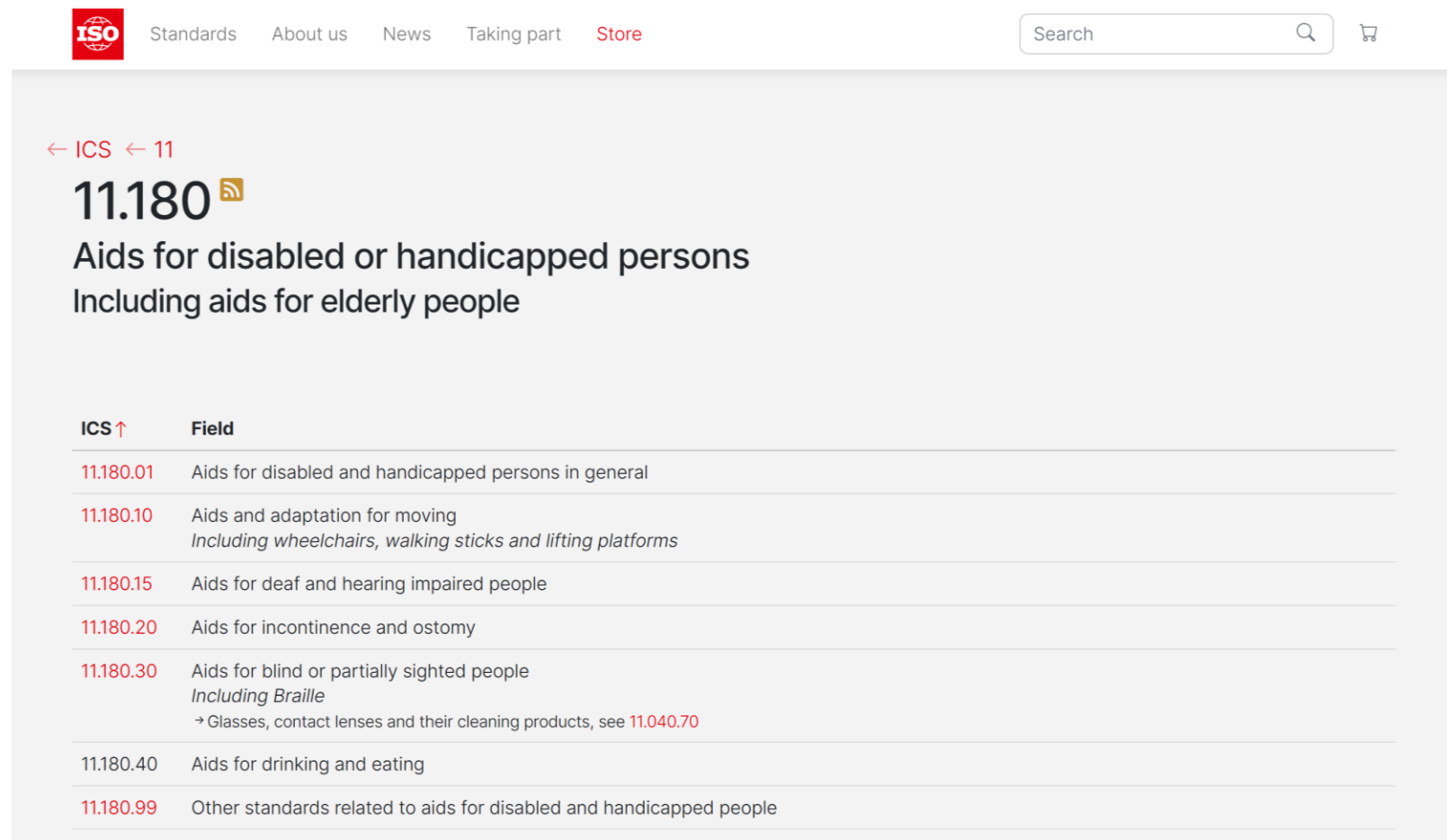
11 

Health care technology

ICS ↑	Field
11.020	Medical sciences and health care facilities in general → Forensic science, see 07.140
11.040	Medical equipment
11.060	Dentistry
11.080	Sterilization and disinfection
11.100	Laboratory medicine
11.120	Pharmaceutics
11.140	Hospital equipment <i>Including hospital beds, surgical tables, medical garments, medical gloves, containers for sharp disposal, etc.</i>
11.160	First aid <i>Including kits, equipment, facilities and medical transport for first aid</i>
11.180	Aids for disabled or handicapped persons <i>Including aids for elderly people</i>

ISO 9999:2022
11.180

ISO Classification of Assistive Technology



The screenshot shows the ISO website's navigation bar with links for Standards, About us, News, Taking part, and Store. A search bar is present on the right. Below the navigation bar, the breadcrumb trail reads "← ICS ← 11". The main heading is "11.180" with a small icon. Below this, the text reads "Aids for disabled or handicapped persons" and "Including aids for elderly people". A table lists the ICS codes and their corresponding fields:

ICS ↑	Field
11.180.01	Aids for disabled and handicapped persons in general
11.180.10	Aids and adaptation for moving <i>Including wheelchairs, walking sticks and lifting platforms</i>
11.180.15	Aids for deaf and hearing impaired people
11.180.20	Aids for incontinence and ostomy
11.180.30	Aids for blind or partially sighted people <i>Including Braille</i> → Glasses, contact lenses and their cleaning products, see 11.040.70
11.180.40	Aids for drinking and eating
11.180.99	Other standards related to aids for disabled and handicapped people

ISO 9999:2022
11.180.xx

Enabling environments

Box 6.1. Definitions and concepts

Accessibility – in common language, the ability to reach, understand, or approach something or someone. In laws and standards on accessibility, it refers to what the law requires for compliance.

Universal design – a process that increases usability, safety, health, and social participation, through design and operation of environments, products, and systems in response to the diversity of people and abilities (1).

Usability, though, is not the only goal of universal design, and “adaption and specialized design” are a part of providing customization and choice, which may be essential for addressing diversity. Other overlapping terms for the same general concept are “design for all” and “inclusive design”.

Standard – a level of quality accepted as a norm. Sometimes standards are codified in documents such as “guidelines” or “regulations”, both with specific definitions, with different legal implications in different legal systems. An example is Part M of the Building Regulations in the United Kingdom of Great Britain and Northern Ireland. Standards can be voluntary or compulsory.

Public accommodations – buildings open to and provided for the public, whether publicly owned (such as courts, hospitals, and schools) or privately owned (such as shops, restaurants, and sports stadia) as well as public roads.

Transportation – vehicles, stations, public transportation systems, infrastructure, and pedestrian environments.

Communication – “includes languages, text displays, Braille, tactile communication, large print, and accessible multimedia as well as written, audio, plain-language, human-reader and augmentative and alternative modes, means, and formats of communication, including accessible information and communication technology” (2). These formats, modes, and means of communication may be physical, but are increasingly electronic.



Impact of assistive technology

Box 4.3. Money well spent: The effectiveness and value of housing adaptations

Public spending on housing adaptations for people with difficulties in functioning in the United Kingdom of Great Britain and Northern Ireland amounted to more than £220 million in 1995, and both the number of demands and unit costs are growing. A 2000 research study examined the effectiveness of adaptations in England and Wales, using interviews with recipients of major adaptations, postal questionnaires returned by recipients of minor adaptations, administrative records, and the views of visiting professionals. The main measure of “effectiveness” was the degree to which the problems experienced by the respondent before adaptation were overcome by the adaptation, without causing new problems. The study found that:

- Minor adaptations (ramps, over-bath showers, and door entry systems, for example) – most costing less than £500 – produced a range of lasting, positive consequences for virtually all recipients: 62% of respondents suggested they felt safer from the risk of accident, and 77% perceived a positive effect on their health.
- Major adaptations (bathroom conversions, extensions, lifts, for example) in most cases had transformed people’s lives. Before adaptations, people used words like “prisoner”, “degraded”, and “afraid” to describe their situations; following adaptations, they spoke of themselves as “independent”, “useful”, and “confident”.
- Where major adaptations failed, it was typically because of weaknesses in the original specification. Adaptations for children sometimes failed to allow for the child’s growth, for example. In other cases, policies intended to save money resulted in major waste. Examples included extensions that were too small or too cold to use, and cheap but ineffective substitutes for proper bathing facilities.
- The evidence from recipients suggests that successful adaptations keep people out of hospitals, reduce strain on carers, and promote social inclusion.
- Benefits were most pronounced where careful consultation with users took place, where the needs of the whole family had been considered, and where the integrity of the home had been respected.

Adaptations appear to be a highly effective use of public resources, justifying investment in health and rehabilitation resources. Further research is needed in diverse contexts and settings.

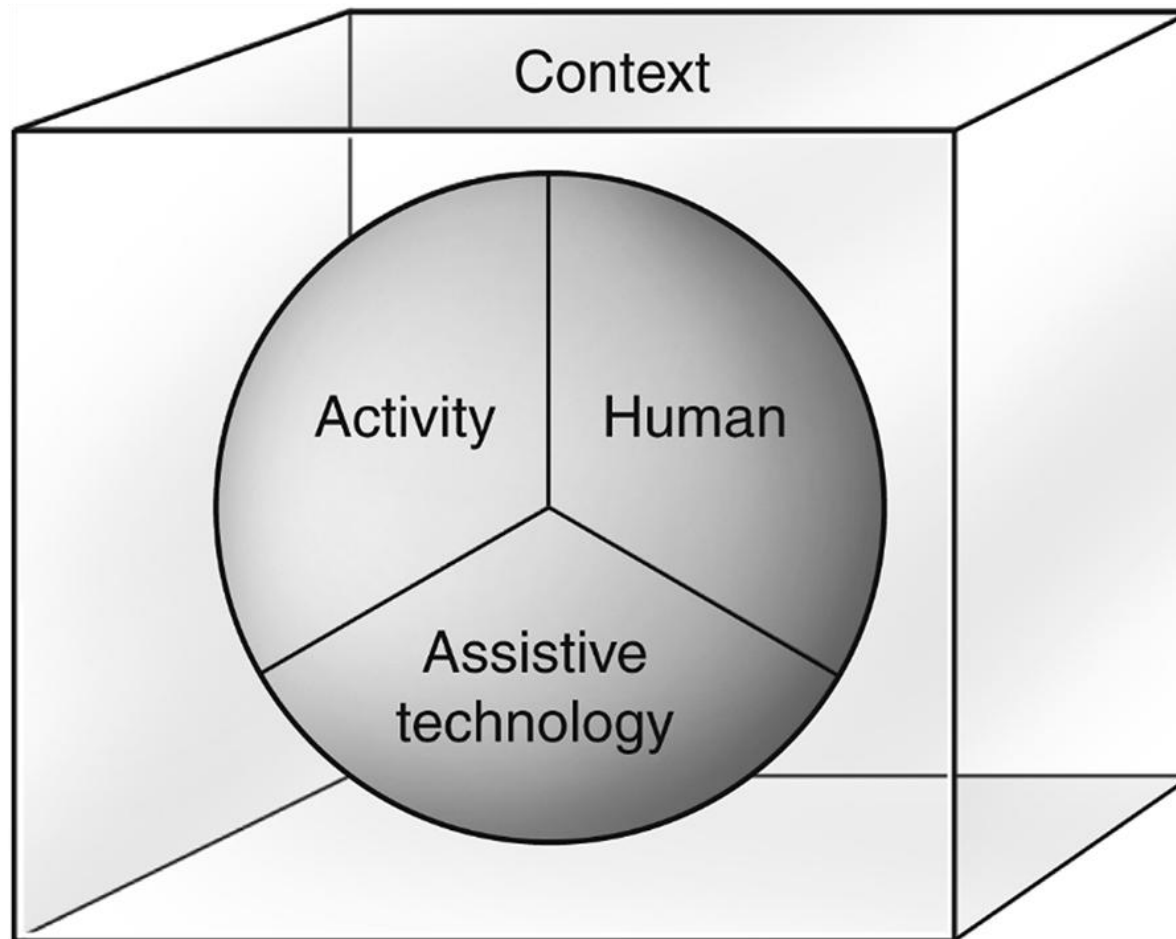
Assistive technology and accessible environments

- Improve the quality of life of citizens with disability
- Increase the productivity of citizens with disability
- Reduce the assistance costs

Assistive technology models

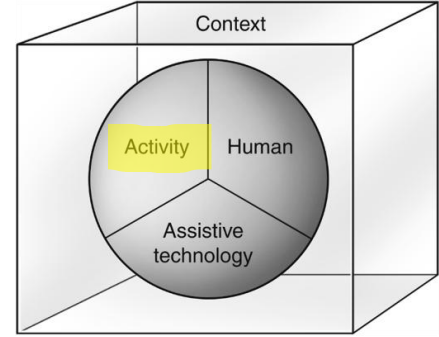
- HEART – Horizontal European Activities in Rehabilitation Technology (1994)
- HAAT – Human-Activity-AssistiveTechnology (1995)
- Enders model (1999)
- Dynamic circles of human activities (2006)

HAAT Model: Human-Activity-Assistive Technology



- The **human** component is the core, with need/wish to perform an **activity**
- Any activity is done in a **context**
- The combination of activity and context outlines the human's required **skills**
- If human's residual skills are not enough for the activity in the context, **assistive technology (AT)** is required
- Since each human's skills are different, any AT device is unique

Activity

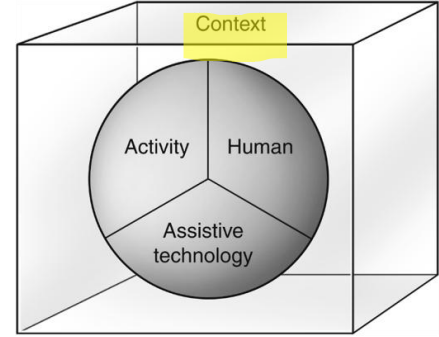


- Fundamental element
- It defines the objective of the assistive device
- It is the functional expression of the human's skills
- It is part of daily life, it can be learned, it is influenced by social and cultural factors related to the context

Activity classification

- Activities of daily life (ADLs):
 - Dressing, hygiene, eating, communication
- Work and productive activities:
 - Home management, educational, vocational, care-of-others activities
- Leisure activities

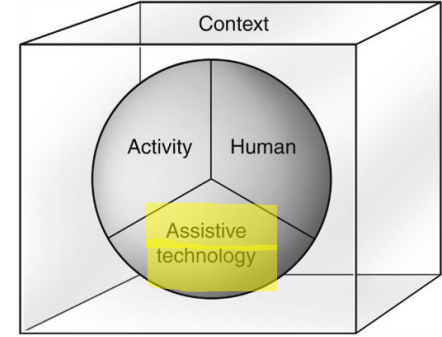
Context



Context description consists of 4 main attributes:

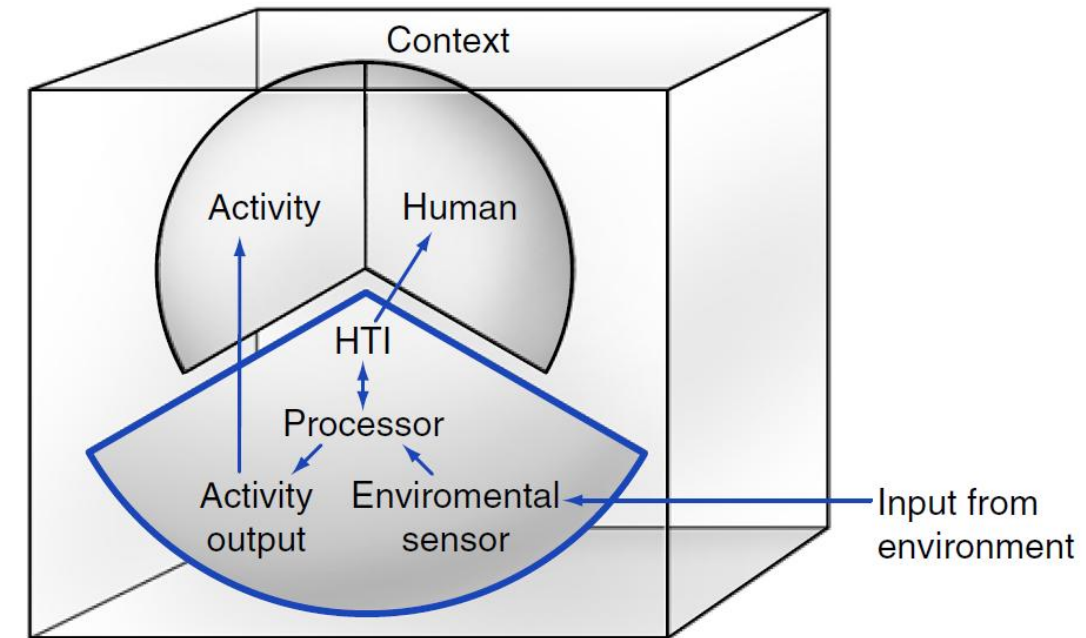
1. **Physical context:** elements of natural and built environments that support or hinder participation and AT use; it includes physical parameters like noise, light, temperature
2. **Social context:** individuals in the environment who affect active participation and AT use, directly or indirectly
3. **Cultural context:** systems of shared meanings that include beliefs, rituals, and values that are broadly held and do not change as quickly as socially held attributes and practices
4. **Institutional context:**
 1. Legislation and related regulations
 2. Policies and funding

Assistive Technology (AT)

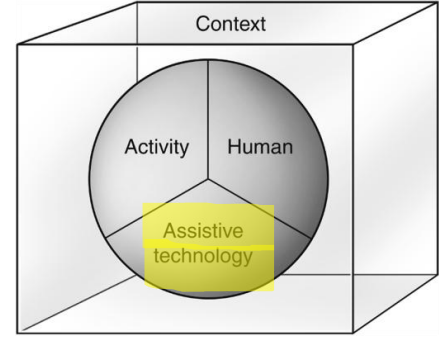


Any AT device consists of 4 main components:

1. **Human-AT interface (HTI)**: it can be physical or cognitive, it is bi-directional
2. **Activity output**: it contributes to the functional performance
3. **Processor**: it translates physical (forces) and cognitive (information) inputs into signals used to control the activity output
4. **Environmental sensor**: it detects information on the context and transfers them to the processing unit and to the human, through the human-AT interface

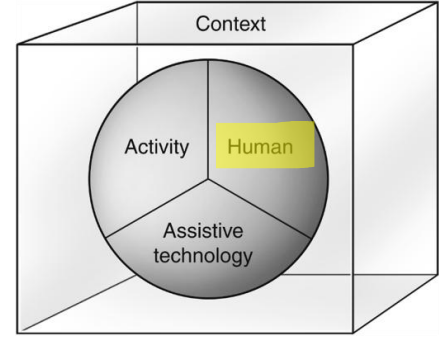


Assistive Technology (AT)



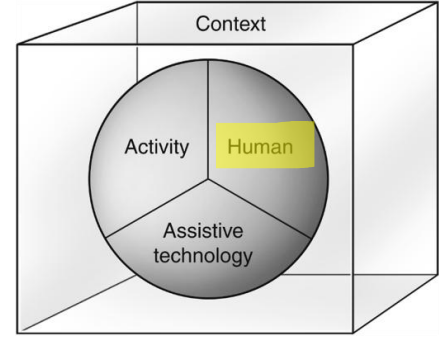
- Assistive technology **enables and promotes inclusion and participation**, especially of persons with disability, aging populations, and people with non-communicable diseases. The primary purpose of assistive products is **to maintain or improve an individual's functioning and independence**, thereby promoting their well-being. They enable people to live healthy, productive, independent and dignified lives, and to participate in education, the labour market and civic life.
- WHO estimates that today **2.5 billion people** need one or more assistive products such as wheelchairs, hearing aids, or apps that support communication and cognition. With a global ageing population and a rise in noncommunicable diseases, this number will rise beyond 3.5 billion by 2050, with many older people needing two or more products as they age.
- Examples of assistive products include hearing aids, wheelchairs, spectacles, prostheses and devices that support memory, among many others. While supporting **independence and well-being**, these products can also help to **prevent or reduce the effects of secondary health conditions**, such as lower limb amputation in people with diabetes. They can also **reduce the need and impact on carers** and mitigate the need for formal health and support services. Moreover, access to appropriate assistive products can have a tremendous impact on **community development and economic growth**.

Human component



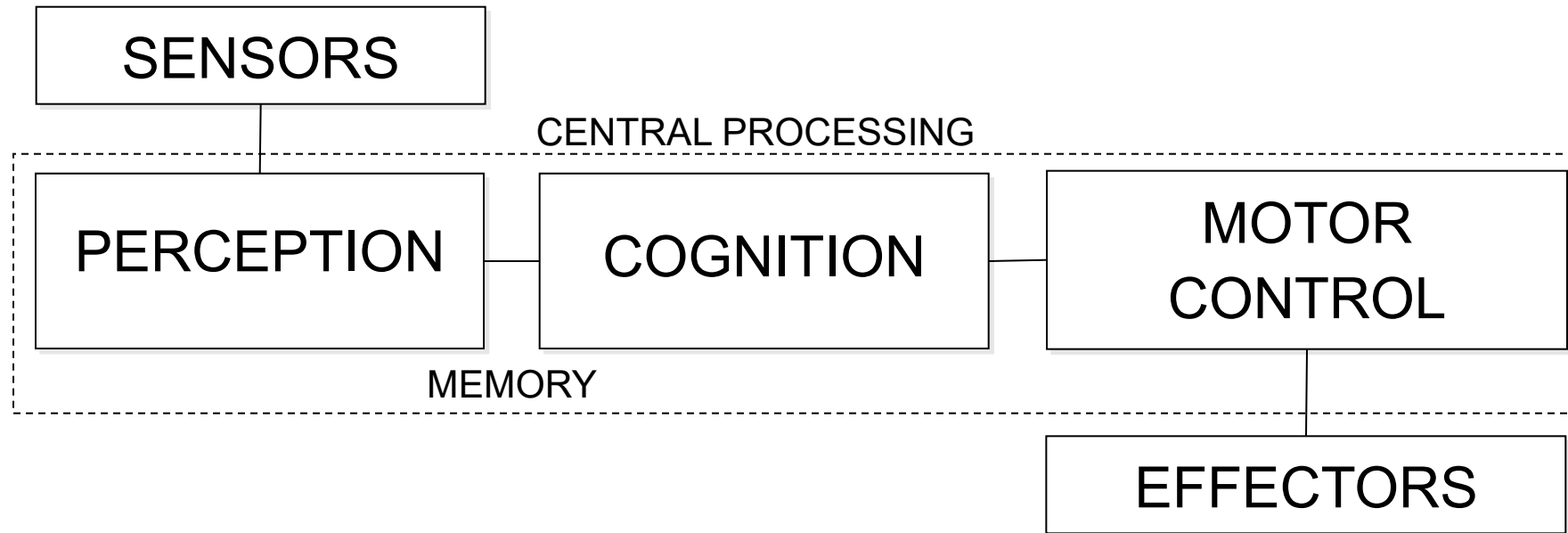
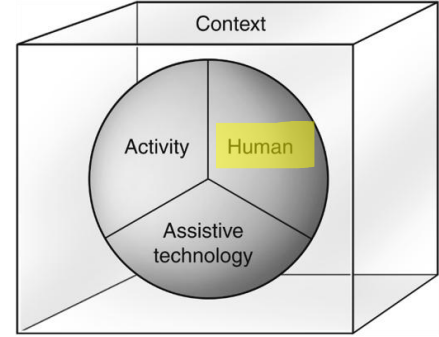
- The person with disabilities, with residual physical and cognitive abilities and emotional states
- He/she selects the activity to perform and the context
- Body functions and structures of ICF are included in the HAAT human component

Human abilities and skills

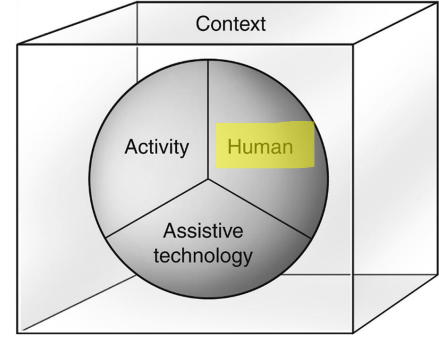


- **Ability** is a person's characteristic, visible in task accomplishment
- **Skill** is a specific expertise level acquired with practice
- **Residual abilities** can be assessed, while it is difficult to predict the skills a person can reach

A model of the human component

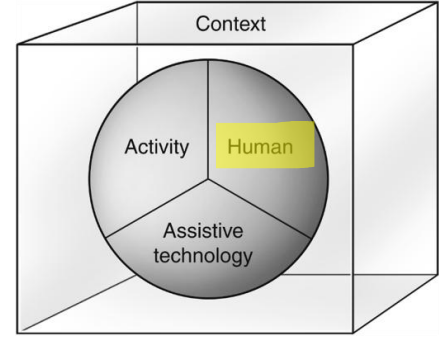


Sensors



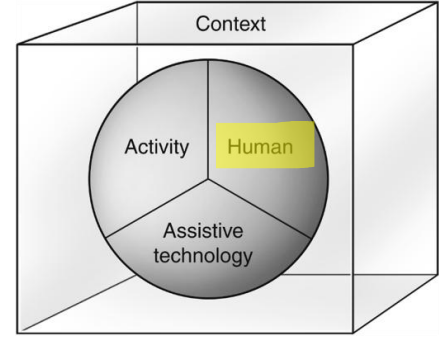
- Inputs from the environment
- Possible limitations:
 - Sensitiveness (minimum detectable threshold)
 - Magnitude (detectable variation in size/magnitude)

Effectors



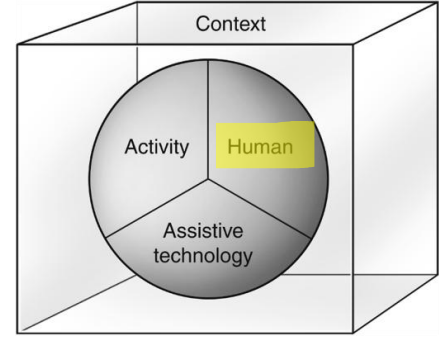
- Neural, muscular and skeletal elements of the human body that provide movement or motor output
- They operate under control from central processing and in response to sensor inputs
- Possible limitations:
 - Impairment of one or a combination of elements

Central processing



- Perception, cognition, neuro-motor control, psychological factors
- Perception: interpretation and understanding of data received from sensors, with use of information contained in the memory
- Cognition: attention, memory, problem solving, decision, learning, language and other functions
- Motor control: integration of sensory, motor and cognitive components in a motor action executed by the effectors
 - Motor planning: movements executed for accomplishing a task
 - Motor learning: acquisition of the skills for executing a task autonomously through practice
- Psycho-social functions: identity, self-protection and motivation

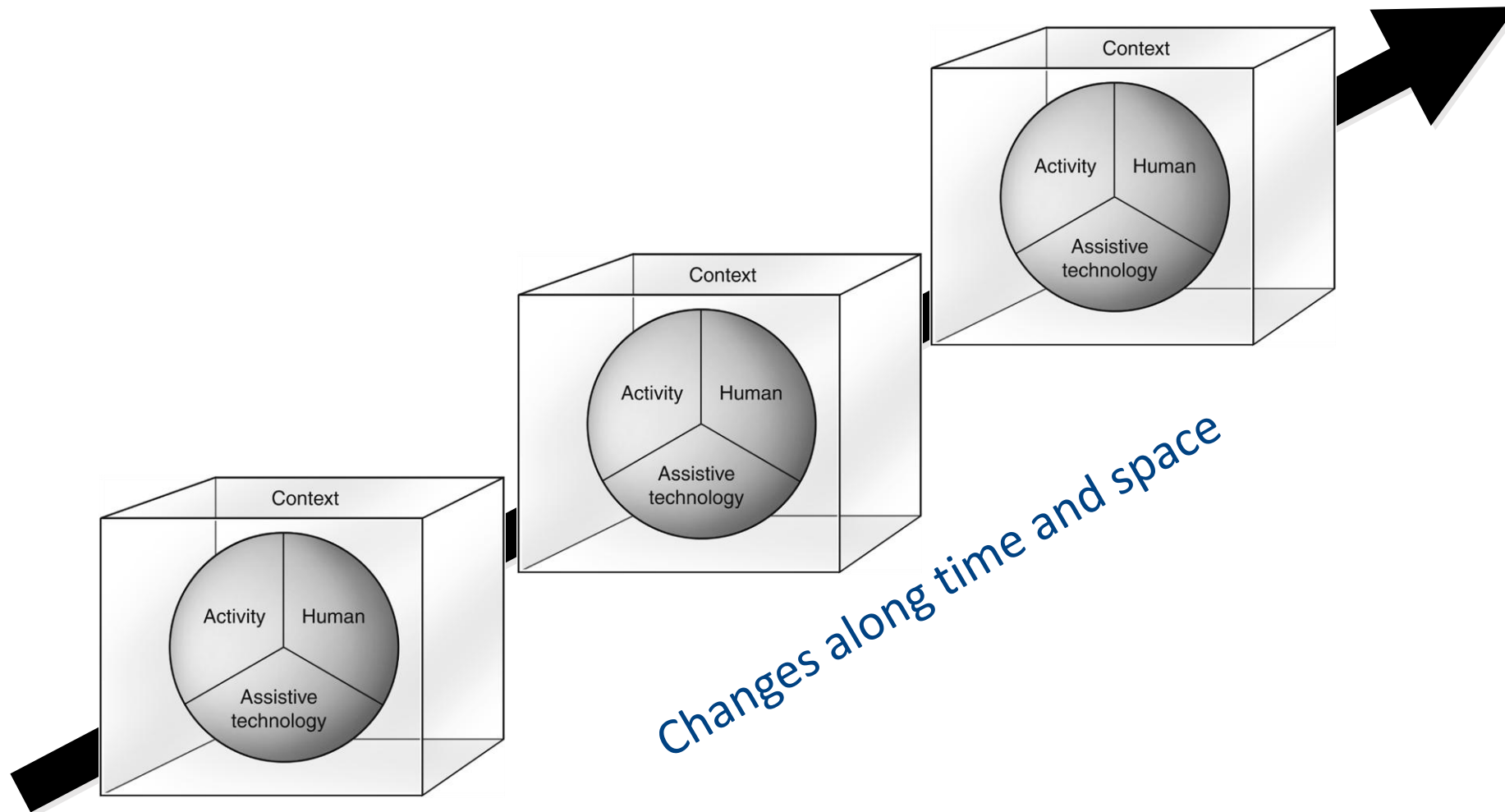
The human component in AT design



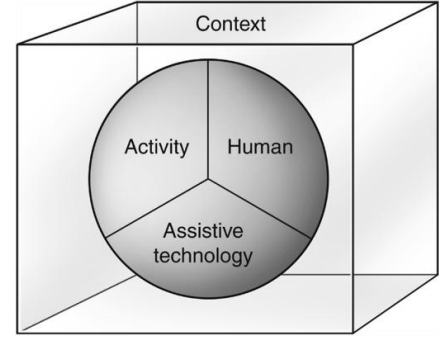
For designing AT, the designer needs to know:

1. What the human can do (abilities and skills)
2. What the human cannot do (limitations)
3. What the human wants to do (motivations)
 - Motivations are directly connected to the human's objectives and needs and to how much an AT device responds to them
 - Abilities and limitations has to be considered with respect to the HAAT model areas

Time and space effects



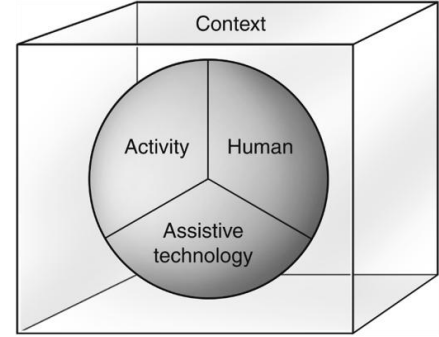
Application of the HAAT model



Four primary applications:

1. Product research and development
2. Product usability studies
3. Clinical assessment
4. Outcome evaluation

Product research and development

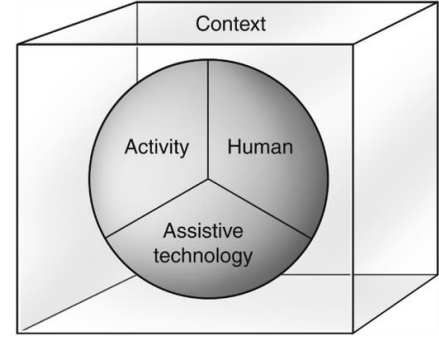


Example: toileting system for children with seating needs

1. Parents of young children providing their opinion on toileting needs, including how they needed to be able to use a toileting product (e.g., it needed to be easily removed from a toilet so others in the household could use the toilet).
2. These opinions were then translated into design features that led to the development of an initial prototype and an evaluation instrument that could be used to critique the prototype.
3. The prototype, along with other currently available products, was then evaluated by parents using the criteria that had been developed in the initial phase.
4. Concurrently, the children who might benefit from the device were seated in it so that an experienced physiotherapist could evaluate the positioning and the children could indicate their opinion of the device.
5. The final stage of the development of the product, before the technology was transferred, involved use of the device in the home over an extended timeframe so the parents and children could provide feedback on the utility and usability of the device
6. The extensive feedback obtained at each stage and the identification of key aspects of the activity, human user, and the contexts in which the device was used resulted in a product that ultimately met its intended outcome.

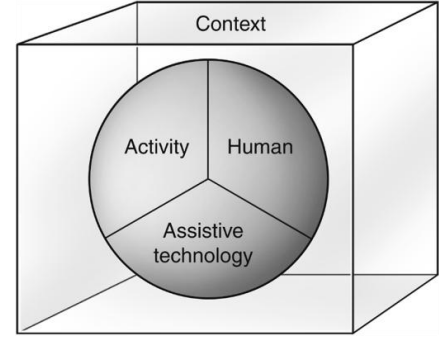
All components of the HAAT model have been considered.

Usability study



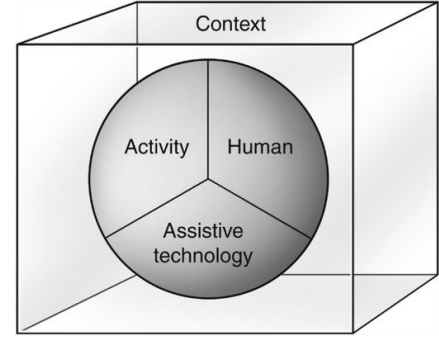
- While **utility** describes how well the device meets its intended function, **usability** describes how well the user is able to access the device's functionality
- Key features of usability:
 1. **Learnability**: All functions of the device are easily learned.
 2. **Efficiency**: The user meets the intended goals of device use in a reasonable amount of time with minimal frustration and effort.
 3. **Memorability**: How the device is used can be easily remembered, particularly when a function has not been used over a long period of time.
 4. **Errors**: This aspect refers to incorrect actions the user makes or actions that are omitted that limit or prevent a device from functioning as intended. It is important that errors can be recognized, that the effect of an error is minimal, that feedback is given to signal an error is made, and that the user can repair an error made.
 5. **Satisfaction**: The user has a positive experience when using the device, which we interpret as satisfaction with how the device functions as well as the image that the device use conveys to the user.
 6. **Ease of use**: This last criterion involves many of the previously mentioned ideas, but it also makes explicit the idea that a device must be simple to use on a regular basis, which we interpret as minimizing the number of steps required to generate the desired output.
- Two major perspectives of usability analysis are described:
 - In the first instance, the goal is to identify and rectify problems that a person encounters when using the device.
 - The second involves performing a number of tasks with the device and measuring the user's performance.
- Although both will provide useful information on problems with device use, the second type of analysis provides details on steps necessary to use the device as well as cognitive, communicative, sensory, and physical demands of device use.

Clinical assessment



- Basic process:
 - identification of the need to be addressed by AT use,
 - assessment of key aspects,
 - synthesis of the assessment results,
 - device recommendation.
- Assessment is user centered:
 - the user of AT—and others as appropriate such as a parent, child, or spouse—are equal partners on the assessment team and that their goals for device use are paramount in driving the assessment process.
- Assessment can include both formal and informal instruments:
 - formal instruments: tests developed using rigorous procedures with established measurement properties, administered and interpreted in a standardized manner
 - informal assessment: observation and interview to gather input and information.
- The clinical team includes the AT users (and others, such as family members), audiologists, occupational and physical therapists, speech language pathologists, nurses, physicians, teachers, rehabilitation **engineers**, and rehabilitation and educational assistants.
- Institutional policies and practices determine who is present during an assessment and what their responsibilities are regarding different aspects of the assessment.
- Assessment culminates in device recommendation and acquisition.
- When possible, the assessment process also results in procurement of funding. This component requires knowledge of who is responsible to request funding and what is required to make that request.

Outcome evaluation



- Two aspects:
 - Evaluation of the outcome of device use by an individual client, at several points after acquisition of the device: immediately after acquisition and short- and long-term follow-up.
 - Outcome of device use for a group of individuals.
- Outcome evaluation is based on the goals identified by the client regarding the acquisition of the device. This evaluation determines how well these goals have been met.
- In addition, other outcomes, such as satisfaction with the device and the psychosocial impact of the device use can also be incorporated into the individual client evaluation
- The HAAT model was used to interpret data in a study of children's perceptions of their use of writing and communication aids. The HAAT model was useful to identify barriers and enablers to the use.
- On a larger scale, evaluation of the outcomes of device use with a sample of clients provides information on device use, the ability of the device to meet the intended need, and feedback on device design for future development. Commonly, this type of evaluation is conducted as part of a formal research study with the purpose of building an evidence base related to the AT use. This evidence base supports clinical decisions, influences policy, and justifies new and continued public funding of assistive devices and services.
- The HAAT model and associated process will benefit from continued development of more specific processes and methods to support outcome evaluation research

Universal design

BOX 2.1 Principles of Universal Design

1. Equitable use: The design is useful and marketable to people with diverse abilities.
2. Flexibility in use: The design accommodates a wide range of individual preferences and abilities.
3. Simple and intuitive use: Use of the design is easy to understand regardless of the user's experience, knowledge, language skills, or current concentration level.
4. Perceptible information: The design communicates necessary information effectively to the user regardless of ambient conditions or the user's sensory abilities.
5. Tolerance for error: The design minimizes hazards and the adverse consequences of accidental or unintended actions.
6. Low physical effort: The design can be used efficiently and comfortably and with a minimum of fatigue.
7. Size and space for approach and use: Appropriate size and space is provided for approach, reach, manipulation, and use regardless of user's body size, posture, or mobility.

Erroneously intended as “a product for all”

It is a philosophy that encourages designers to consider the needs of the largest possible range of users

It typically brings to products for the largest possible citizens range, not necessarily for all

- “Universal Design” (in USA and Japan)
- “Design for All” (in Europe)
- “Inclusive Design” (in UK)

- “... designers ensure that their products and services address the needs of the widest possible audience”

Definition by the British Ministry of Trade and Industry, 2000

- “... Universal design is an approach to design that incorporates products as well as building features which, to the greatest extent feasible, can be used by everyone”

Definition by universal design pioneer Ron Mace, 1985

Universal design principles:

1. Equitable use

Principle 1: Equitable Use

The design is useful and marketable to people with diverse abilities.

Guidelines:

- 1a. Provide the same means of use for all users: identical whenever possible; equivalent when not.
- 1b. Avoid segregating or stigmatizing any users.
- 1c. Provisions for privacy, security, and safety should be equally available to all users.
- 1d. Make the design appealing to all users.



Universal design principles:

2. Flexibility in use

Principle 2: Flexibility in Use

The design accommodates a wide range of individual preferences and abilities.

Guidelines:

- 2a. Provide choice in methods of use.
- 2b. Accommodate right- or left-handed access and use.
- 2c. Facilitate the user's accuracy and precision.
- 2d. Provide adaptability to the user's pace.



Adjustable computer table height



Right & left-handed scissors

Universal design principles:

3. Simple and intuitive use

Principle 3: Simple and Intuitive Use

Use of the design is easy to understand, regardless of the user's experience, knowledge, language skills, or current concentration level.

Guidelines:

- 3a. Eliminate unnecessary complexity.
- 3b. Be consistent with user expectations and intuition.
- 3c. Accommodate a wide range of literacy and language skills.
- 3d. Arrange information consistent with its importance.
- 3e. Provide effective prompting and feedback during and after task completion.



Universal design principles:

4. Perceptible information

Principle 4: Perceptible Information

The design communicates necessary information effectively to the user, regardless of ambient conditions or the user's sensory abilities.

Guidelines:

- 4a. Use different modes (pictorial, verbal, tactile) for redundant presentation of essential information.
- 4b. Provide adequate contrast between essential information and its surroundings.
- 4c. Maximize "legibility" of essential information.
- 4d. Differentiate elements in ways that can be described (i.e., make it easy to give instructions or directions).
- 4e. Provide compatibility with a variety of techniques or devices used by people with sensory limitations.

Lighting accentuates train doorways and the adjoining gates prevent riders from falling onto the tracks



Universal design principles:

5. Tolerance for error

Principle 5: Tolerance for Error

The design minimizes hazards and the adverse consequences of accidental or unintended actions.

Guidelines:

- 5a. Arrange elements to minimize hazards and errors: most used elements, most accessible; hazardous elements eliminated, isolated, or shielded.
- 5b. Provide warnings of hazards and errors.
- 5c. Provide fail safe features.
- 5d. Discourage unconscious action in tasks that require vigilance.



Universal design principles:

6. Low physical effort

Principle 6: Low Physical Effort

The design can be used efficiently and comfortably and with a minimum of fatigue.

Guidelines:

- 6a. Allow user to maintain a neutral body position.
- 6b. Use reasonable operating forces.
- 6c. Minimize repetitive actions.
- 6d. Minimize sustained physical effort.



Universal design principles:

7. Size and space for approach and use

Principle 7: Size and Space for Approach and Use

Appropriate size and space is provided for approach, reach, manipulation, and use regardless of user's body size, posture, or mobility.

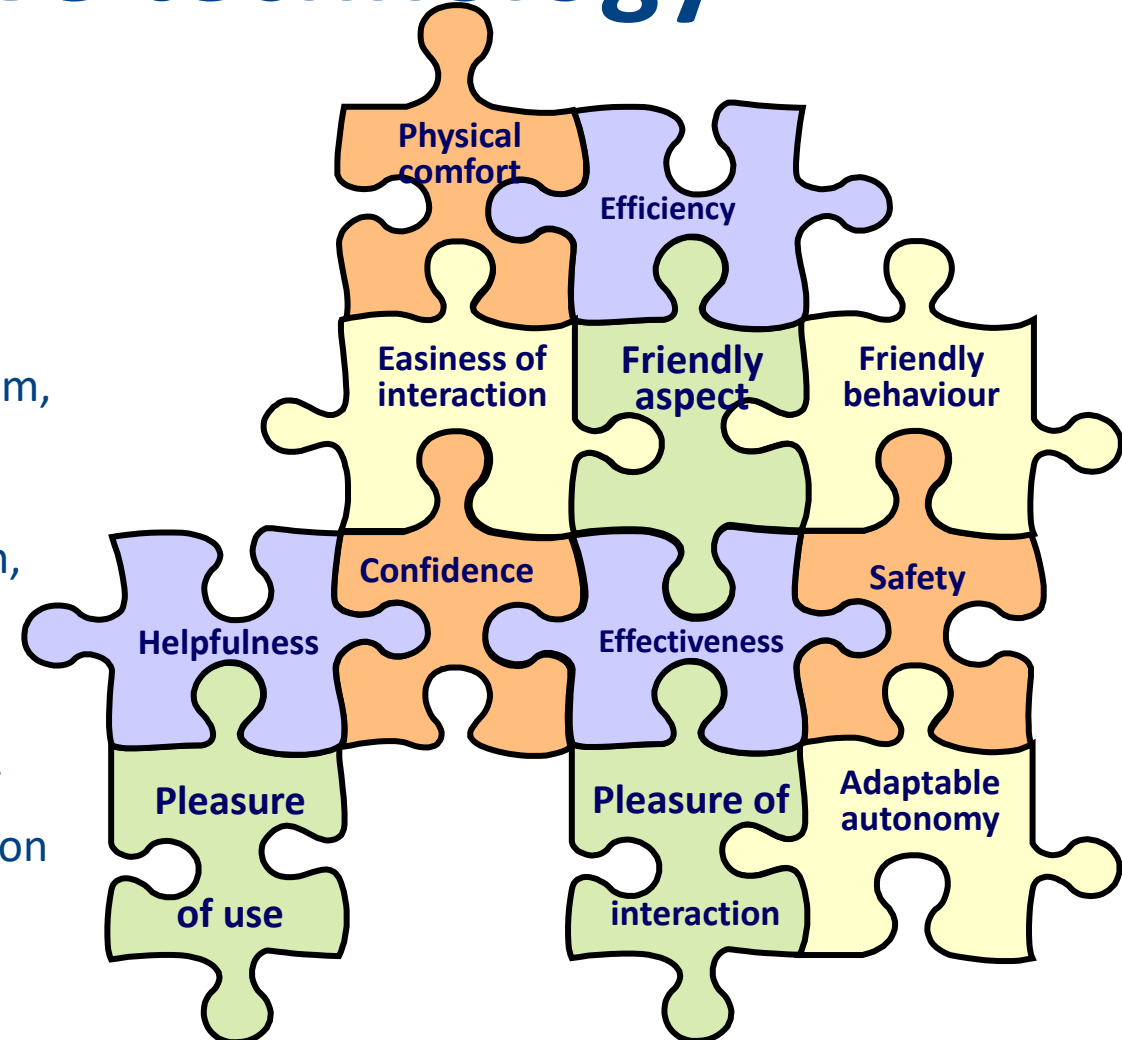
Guidelines:

- 7a. Provide a clear line of sight to important elements for any seated or standing user.
- 7b. Make reach to all components comfortable for any seated or standing user.
- 7c. Accommodate variations in hand and grip size.
- 7d. Provide adequate space for the use of assistive devices or personal assistance.



Acceptability of assistive technology

- **Rational factors:**
perceived helpfulness, effectiveness, efficiency, reliability, robustness
- **Ergonomic factors:**
perceived physical comfort, in the interaction with the system, confidence in contact
- **Emotional factors:**
positive emotional feeling in the interaction with the system, pleasure of use and interaction, liking aspects
- **Psychological factors:**
positive relationship between the user and the robot personalities, proper balance of robot autonomy, respect of user's preferences and privacy, easiness of use and interaction
- **Economic and social factors:**
Cost/benefit ratio, Cost performance ratio, Ethical aspects, Healthcare Policy

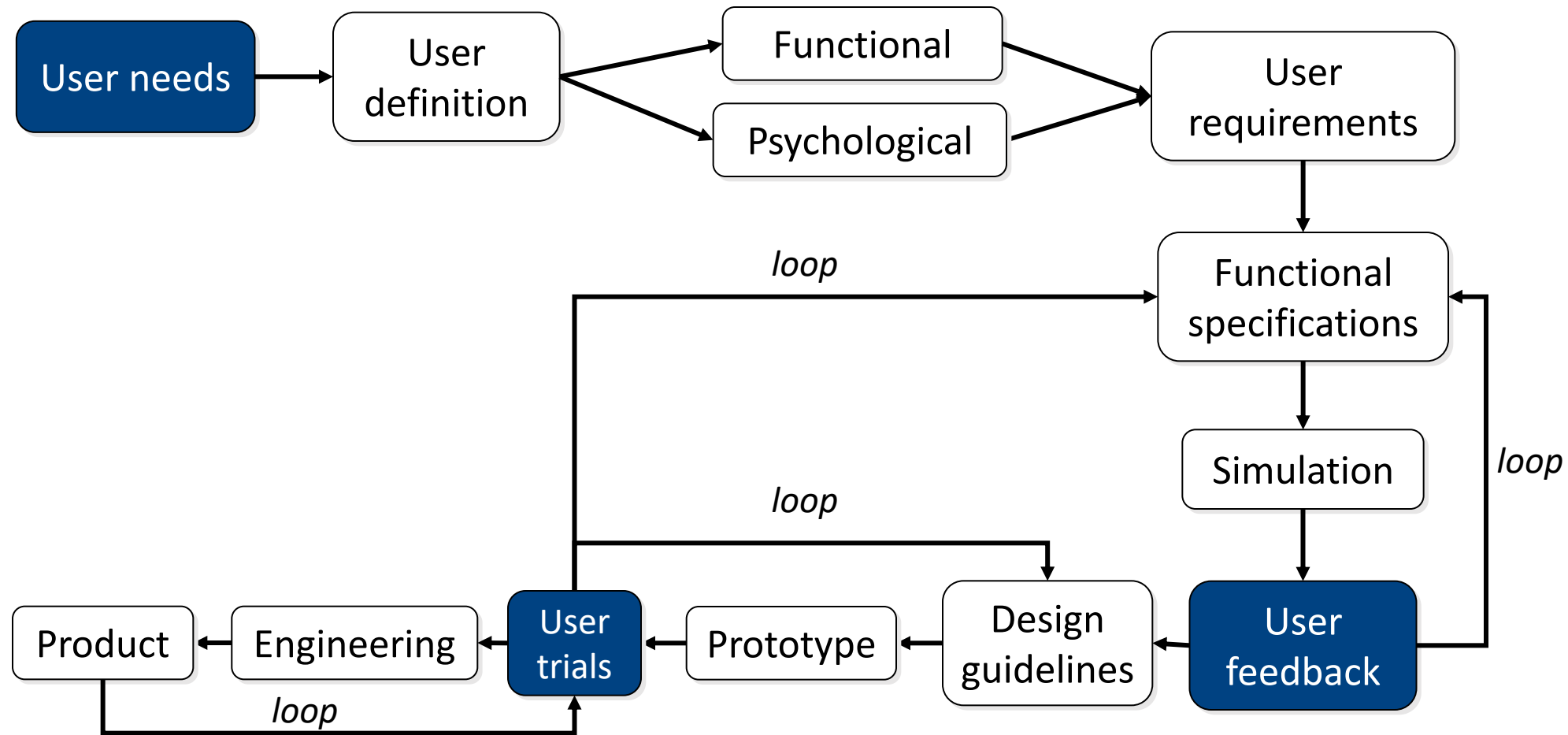


User-Centred Design (UCD)



Involving users in the design process
Users = individuals with disability,
clinical staff, caregivers, family members

User-oriented approach

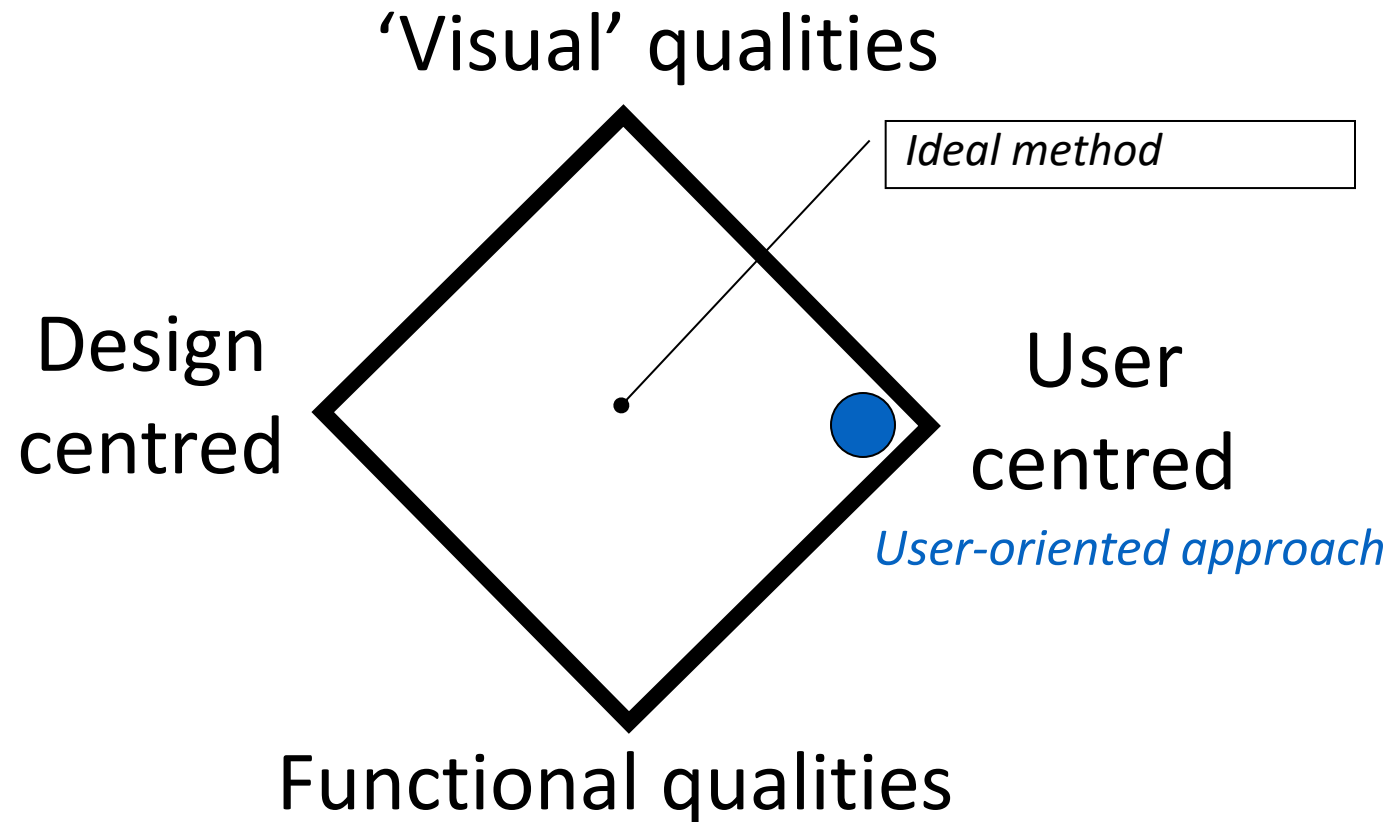




Possible limitations of the user-oriented approach

- Complex, time-demanding
- End users can hardly imagine radically innovative products
- They can only express incremental innovation
- Thus, the user-oriented approach **may limit innovation**

The square of design methods



Double helix design

Design driven

User driven

Technology paradigms

Basic user needs

Inspirational data

Design concepts

*Refinement of concepts
and simulation*

Users' feedback

*Feedback from users on
simulated interaction modalities*

Prototype

System Engineering

User trials

Characterization of Assistive Technologies (AT)

- Assistive vs. rehabilitative or educational technologies
- Low to high tech technology
- Hard technologies and soft technologies
- Appliances vs. tools
- Minimal to maximal technology
- General vs. specific technologies
- Commercial to custom technology

Assistive vs. rehabilitative or educational technologies

- **Assistive technologies** serve a variety of functional needs. They help an individual to carry out a functional activity
- **Rehabilitative** and **educational technologies** are used within a rehabilitative or educational process, respectively, and are used as tools for remediation or rehabilitation, rather than being a part of the person's daily life and functional activities. Often they are used for developing skills for the use of assistive technologies

Low to high tech technology

Not fully precise distinction

- **Low-tech:** simple to make and easy to obtain
 - e.g., simple pencil-and-paper communication boards, modified eating tools, simple splints
- **High-tech:** expensive, more difficult to make and harder to obtain
 - e.g., wheelchairs, electronic communication devices, computer-based aids, assistive robots



Example of low-tech AT



<https://www.youtube.com/watch?v=iyQITEACATk&t=147s>

Example of high-tech AT



Low-tech equivalent



MicroSoft
stock image

<https://www.youtube.com/watch?v=w3RDUxCJWAU&t=52s>

Hard technologies and soft technologies

- **Hard:** tangible, readily available components that can be assembled into assistive technology systems
- **Soft:** human area of decision making, strategies, training, concept formation, and service delivery. They are generally captured as:
 - People
 - Written
 - Computer

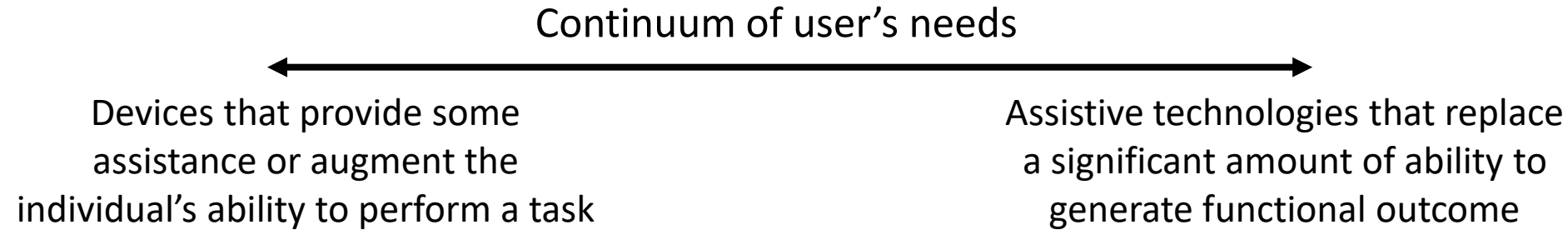
These aspects of technology are much harder to obtain, but without them the hard technologies cannot be successful.

They are difficult to acquire because they are highly dependent on human knowledge rather than on tangible objects

Appliances vs. tools

- **Appliances** provide benefit to an individual independently of the individual's skill level
 - e.g., refrigerator
- **Tools** require the development of skill for their use
 - e.g., hammer

Minimal to maximal technology



- Minimal technologies improve functionality
 - e.g., orthoses, limited-use wheelchairs (e.g., in airports or public spaces)
- Maximal technologies replace functions
 - e.g., prostheses, wheelchairs, communication aids, ...

General vs. specific technologies

- **General-purpose technologies** are used in many different applications
 - e.g., seating and positioning systems, control interfaces, computers
- **Specific technologies** are intended for a specific application
 - e.g., communication devices, wheelchairs, feeding devices, hearing aids, mobility aids for people with visual impairments



MicroSoft
stock images

Commercial to custom technology



Mainstream
technology



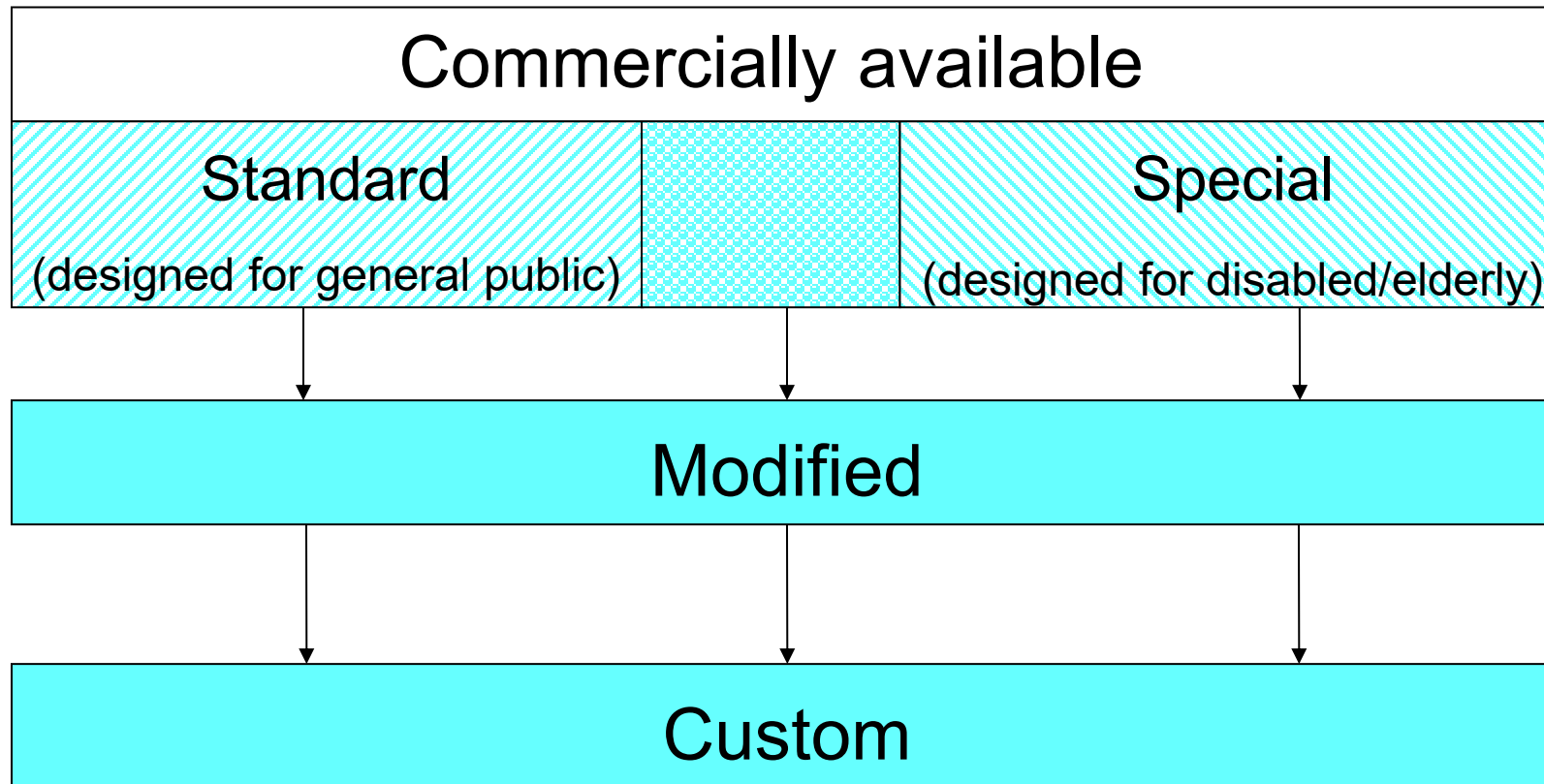
Commercially available
assistive technology



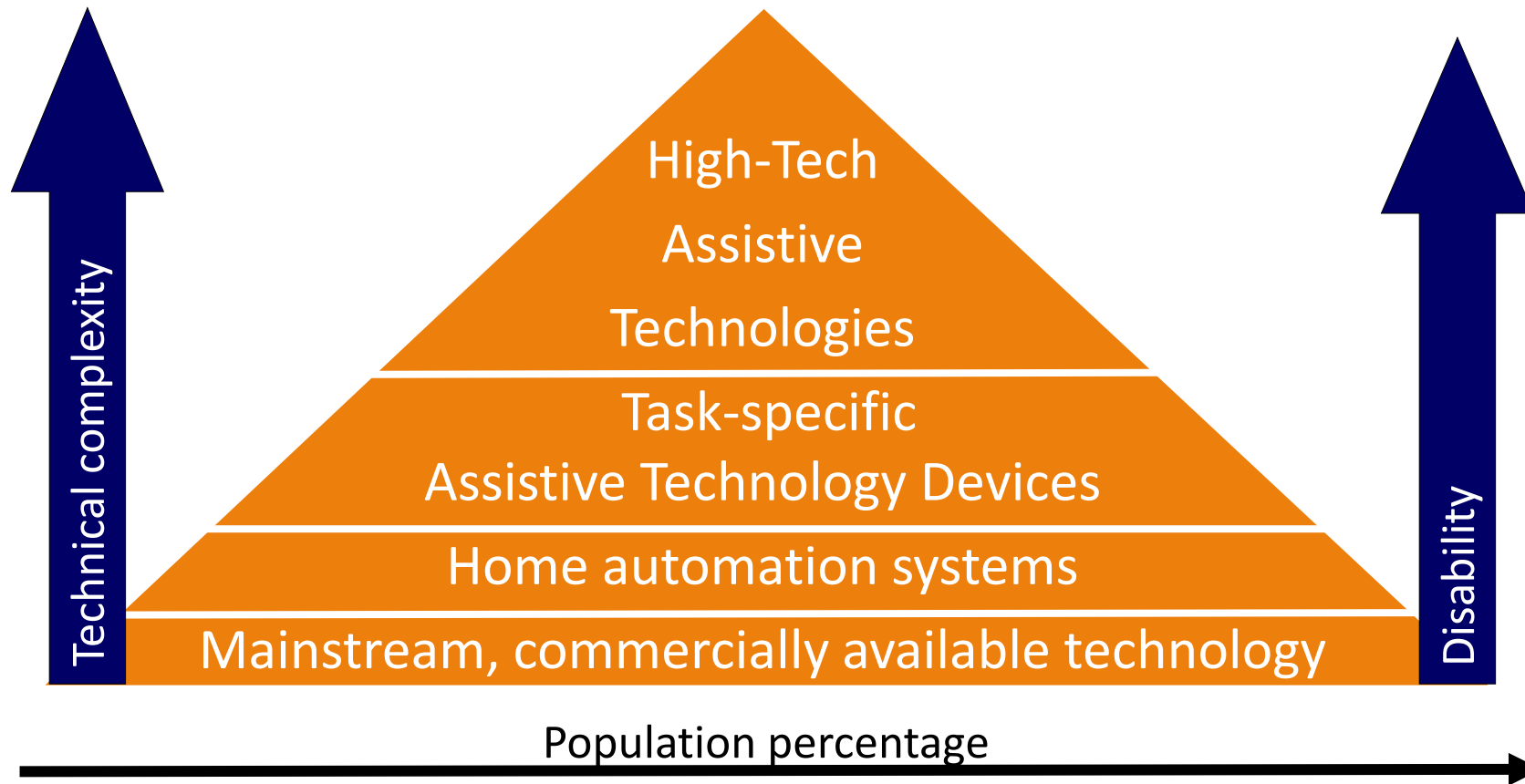
Custom-made
assistive technology

Mass-produced devices are typically easier to obtain and less expensive than devices that are produced specifically for individuals with disabilities

Commercial to custom technology



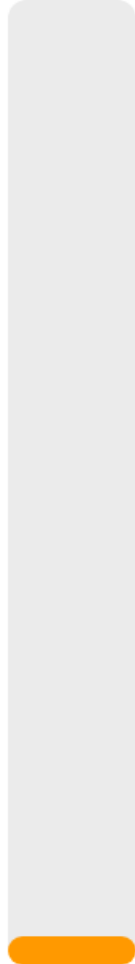
Assistive technology pyramid



Can robots be assistive devices?

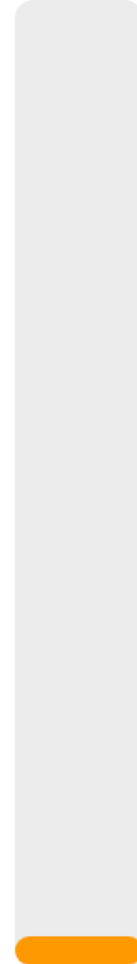


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Yes

0%



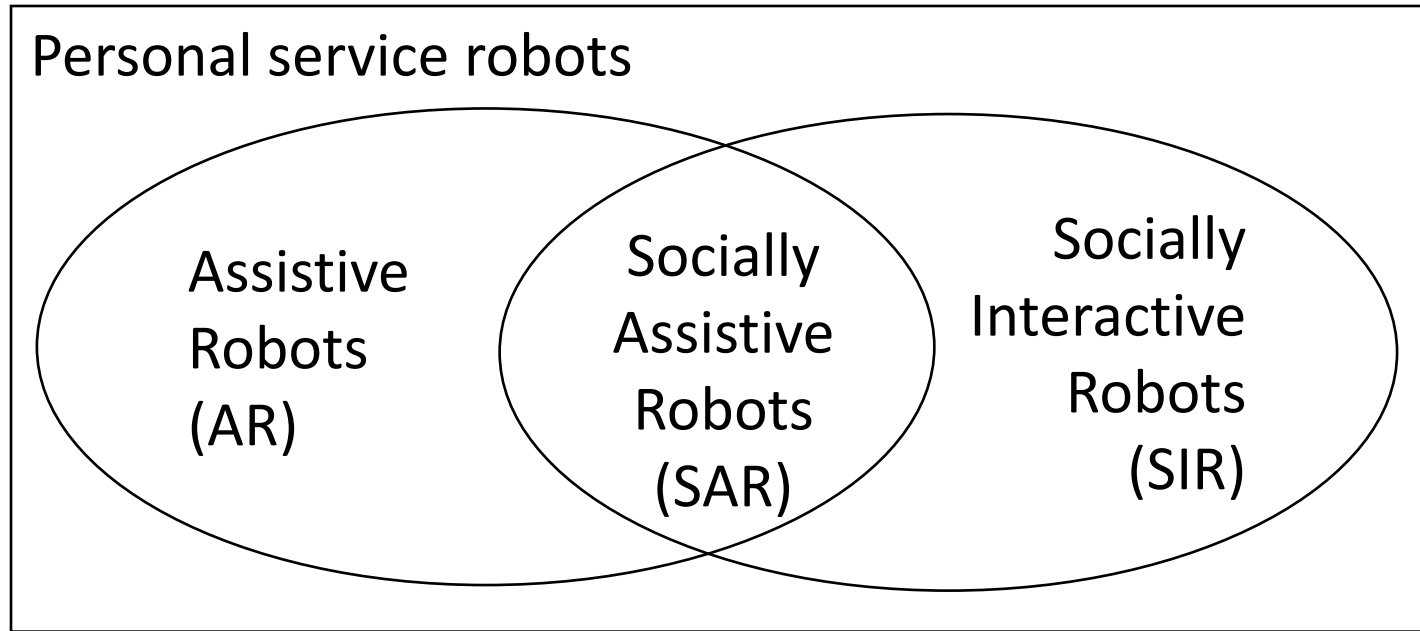
No

Robotics as Assistive Technology (AT)

Robot Classification (ISO 2012)

- **Industrial robot:** “automatically controlled, reprogrammable, multipurpose manipulator, programmable in three or more axes, which can be either fixed in place or mobile for use in industrial automation applications”
- **Service robot:** “robot that performs useful tasks for humans or equipment excluding industrial applications”
- Service robots can be for
 - **Professional** use, used for a commercial task, usually operated by a properly trained operator
 - **Personal** use, used for a non-commercial task, usually by laypersons

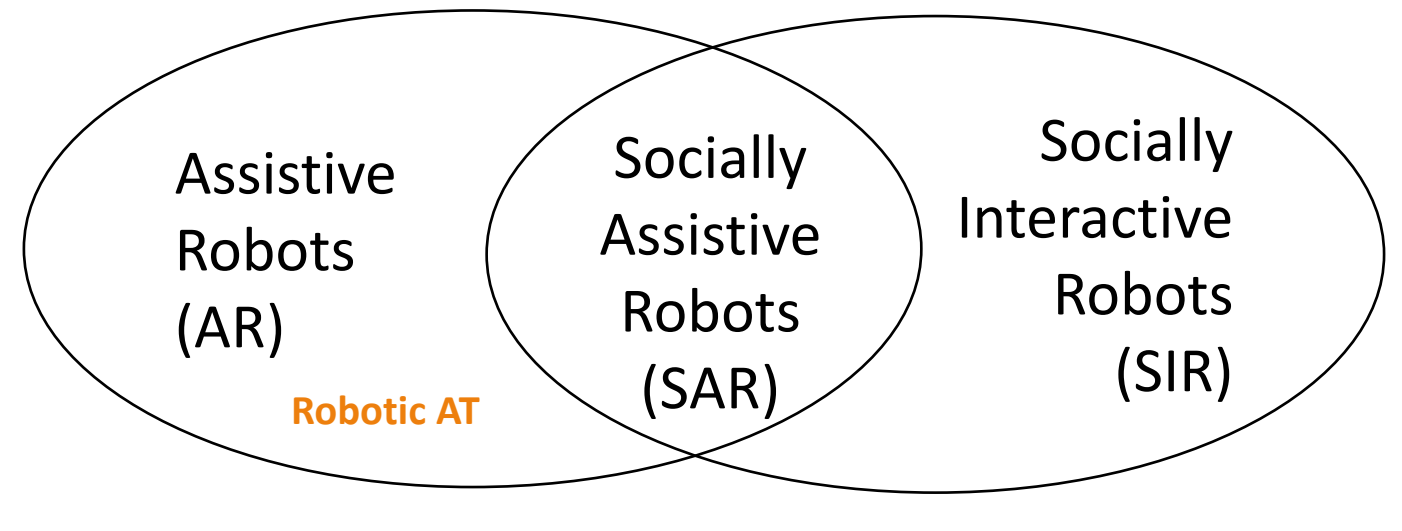
Personal Service Robots



- **Assistive Robots (AR)** give aid or support to a human user
- In **Socially Interactive Robots (SIR)** social interaction plays a key role
- **Socially Assistive Robots (SAR)** assist the user with social interaction
- **Physically Assistive Robots** assist the user with physical interaction

Robotic AT (Assistive Technologies)

Personal service robots



A sub-class of Assistive Robots (AR), because they are used by individuals with disabilities (while AR are used by any user)

- Robotic prosthetic arms and legs
- Robotic manipulators
- Bionic arms
- Exoskeletons



Name a task for a robot to help a person with disability

 0

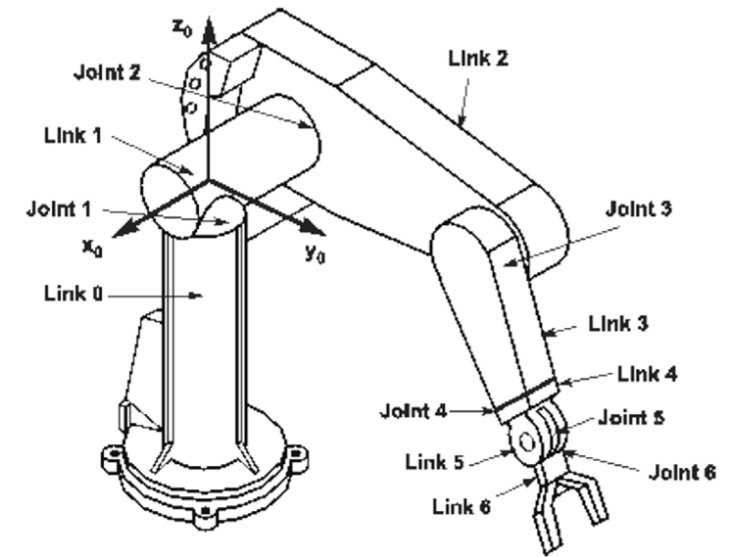
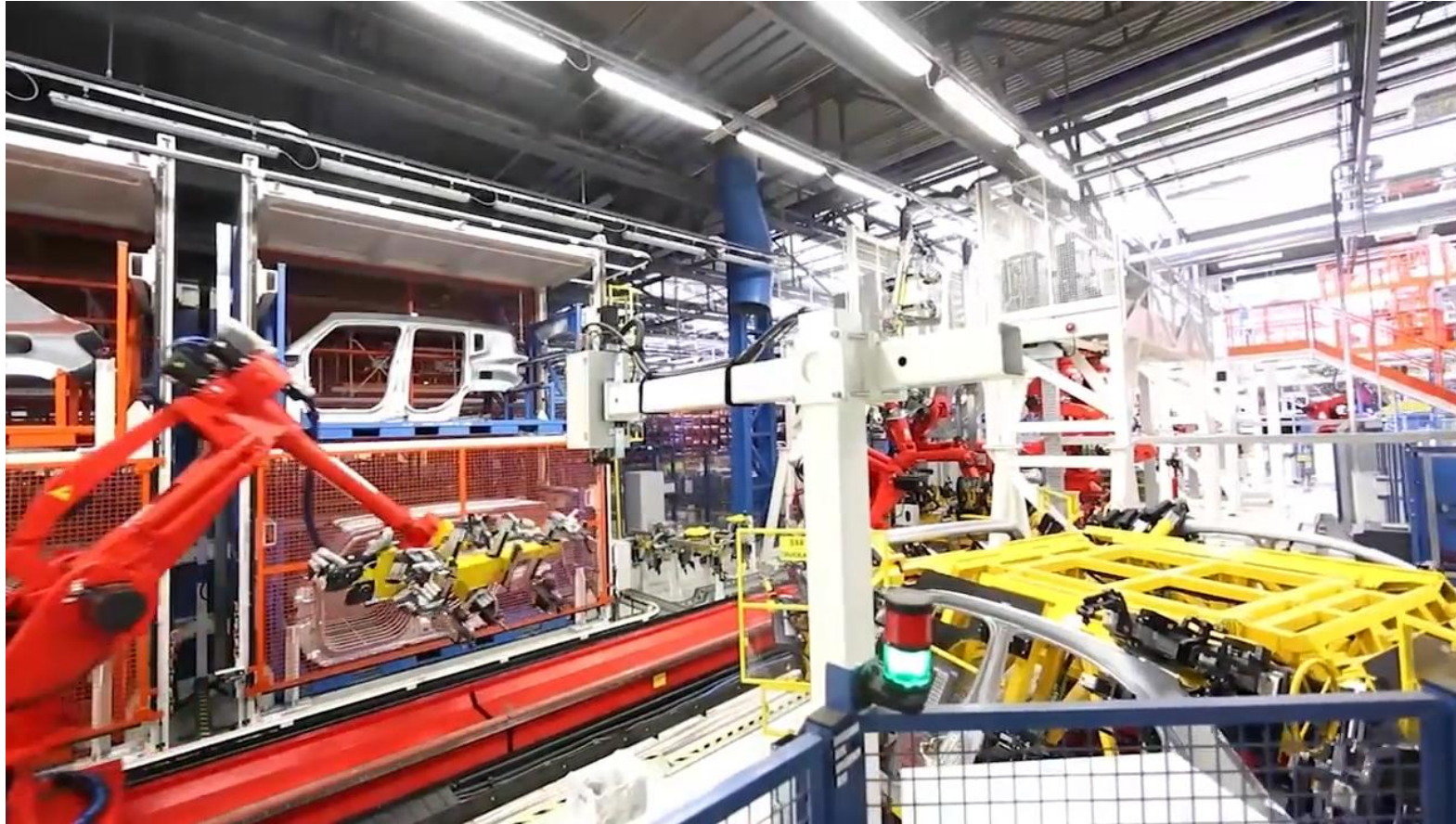
Nobody has responded yet.

Hang tight! Responses are coming in.

Evolution of Robotic Assistive Technologies

Industrial Robots

Structured environments
Robot fixed in the environment



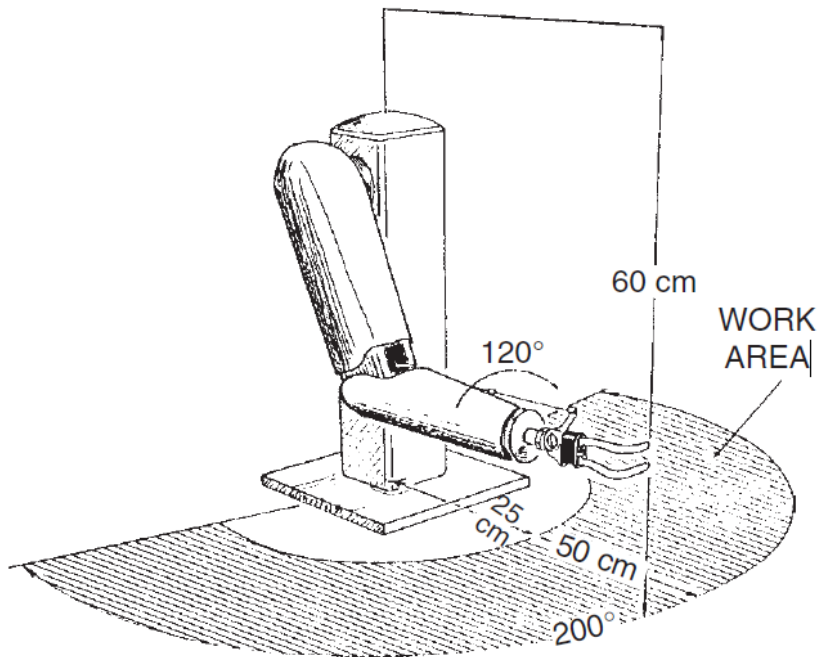
ROBOT

Reliability (minimal requested values):
Mean Time Before Failure = 40,000 hrs
Efficiency $\eta > 99.99875\%$ (Source: COMAU)

Fixed Assistive Robots

Structured environments
Robot fixed in the environment

APL RAWT system



Master-Raid



DeVAR



Track-mounted upside-down PUMA-260 with Otto-Bock gripper, for manipulation of objects in a vocational desktop environment

Service Robots

Robots outside factories need to negotiate **unstructured environments** and react to unexpected situations



- Navigation



- Mobile manipulation

<https://youtu.be/BiemOdLYf48>

Smart Wheelchairs



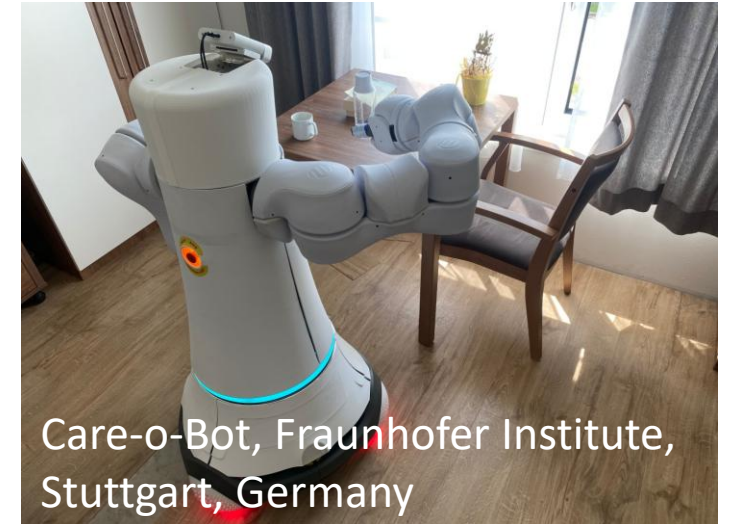
Navigation
sensors and
algorithms



Wheelchair- mounted Manipulators



Mobile Assistive Robots



Care-o-Bot, Fraunhofer Institute,
Stuttgart, Germany



TIDE-MOVAID, 1997,
Scuola Superiore Sant'Anna,
Pisa, Italy



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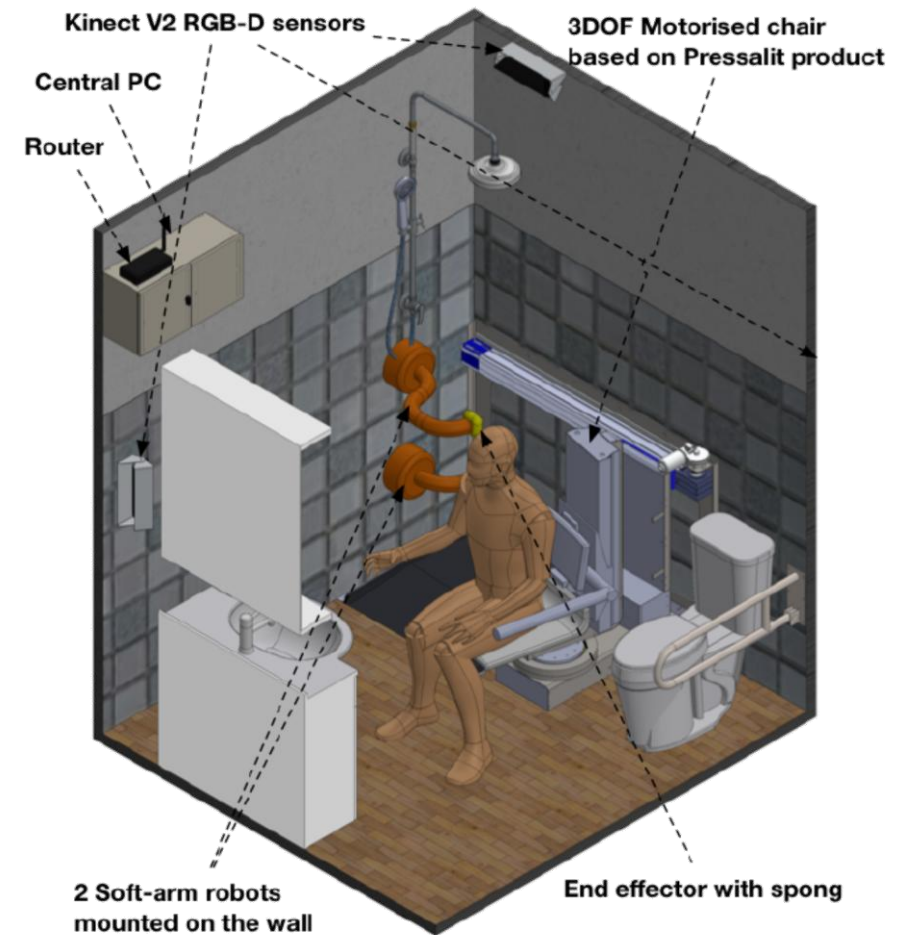
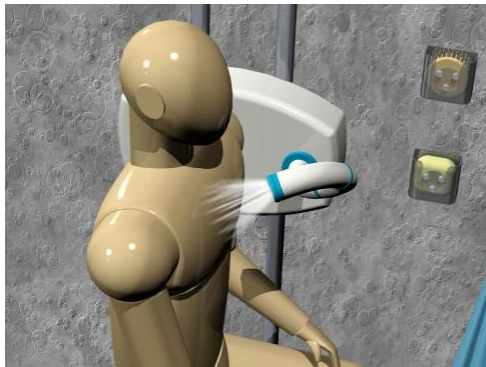
Example of Application of User-centred Design to an Assistive Robot for Showering

I-Support

I-Support: A Service Robotics System for Bathing Tasks

Objectives:

- Autonomy and privacy for the user
- Higher-level assistance for the carer
- A motorized shower chair
- A soft robotic shower arm, for:
 - pouring water and detergent
 - wiping and scrubbing

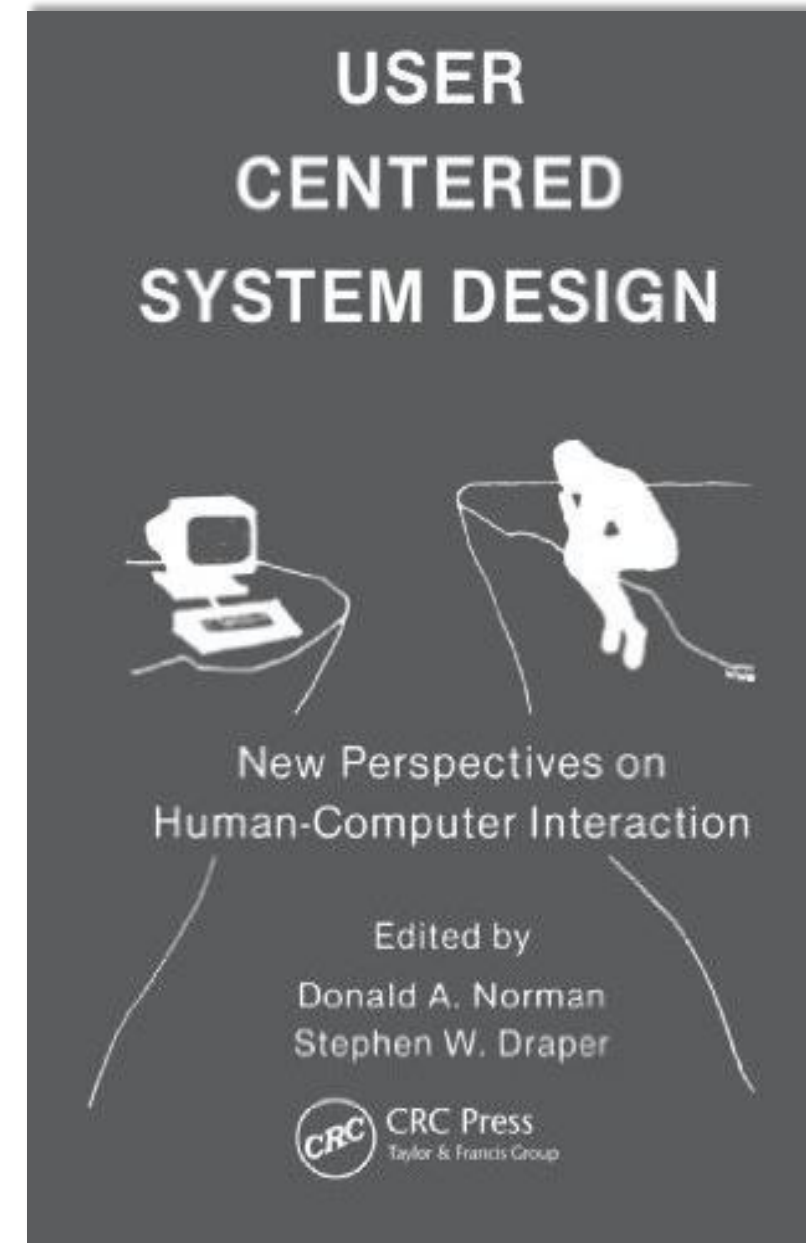


User-Centred Design (UCD)

Definition of the **SCENARIO** and search for **END-USERS**



Definition of **USER NEEDS** through interviews, focus groups, surveys

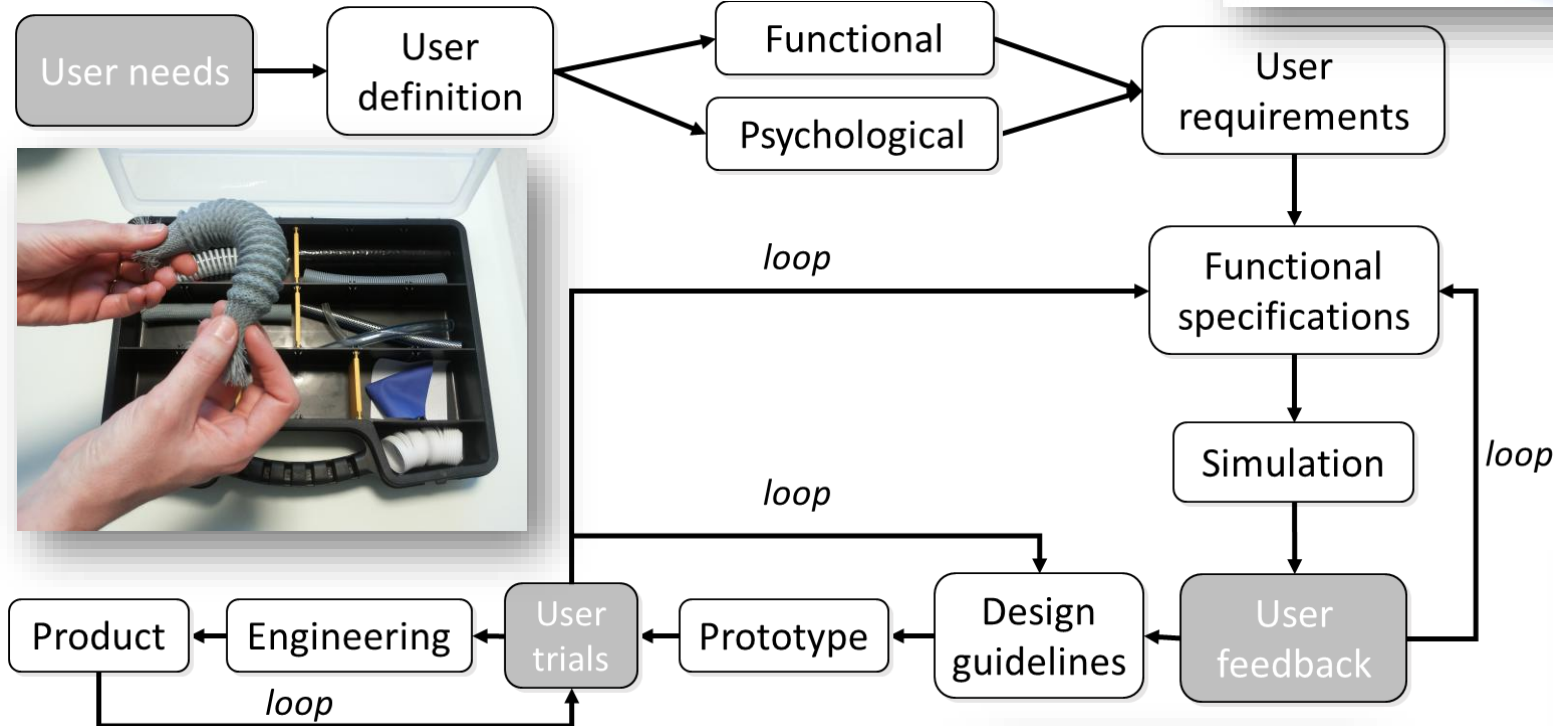
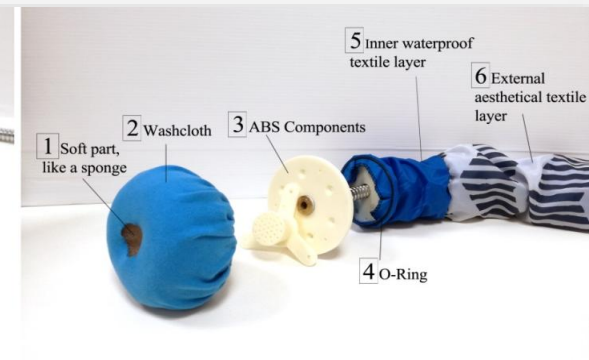


Norman, D. A., & Draper, S. W. (1986). *User centered system design*. Hillsdale, NJ.

User-oriented Design

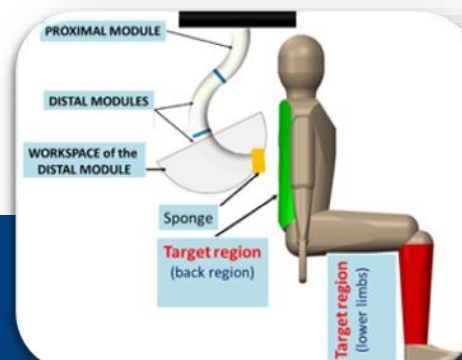
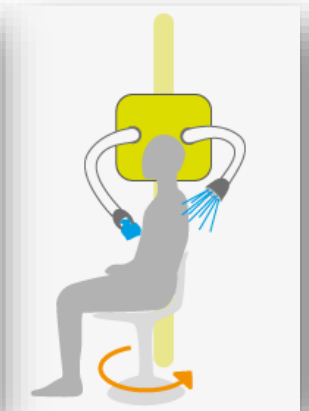
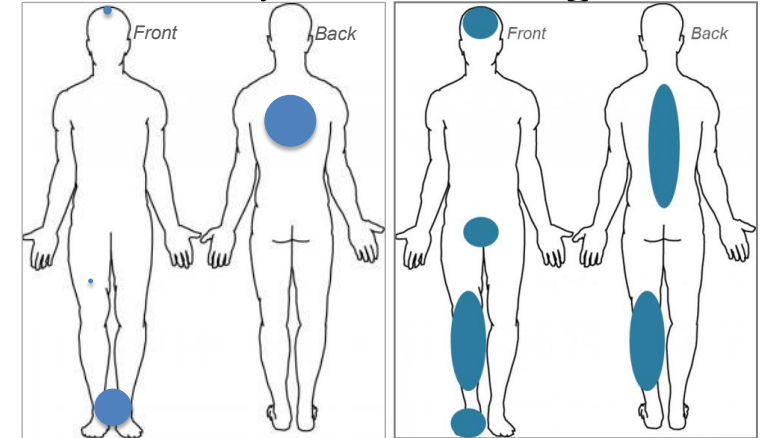
aesthetics

shower curtain fabric

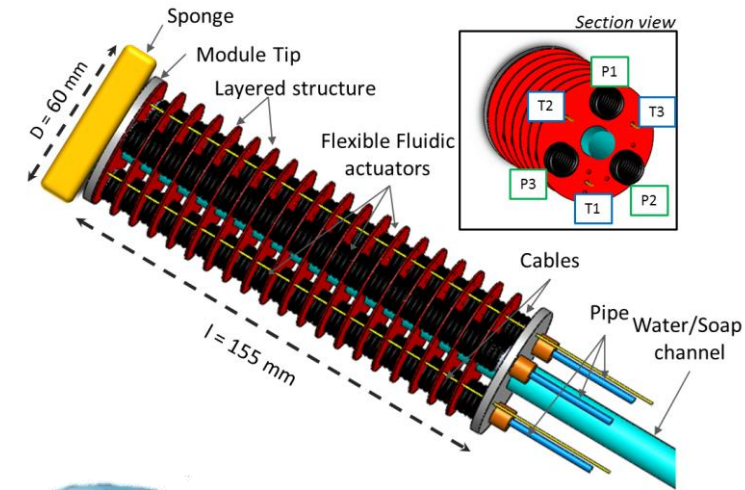


Elderly

Care givers



I-Support Robot Arm



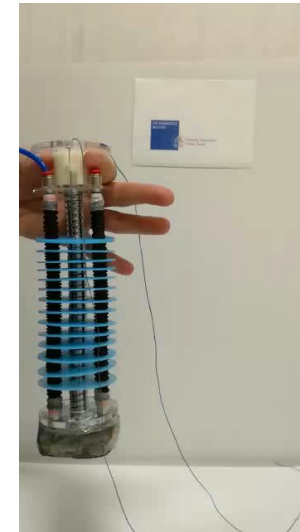
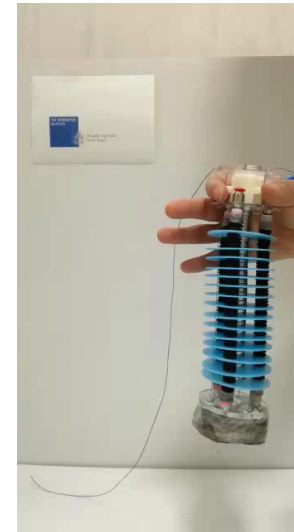
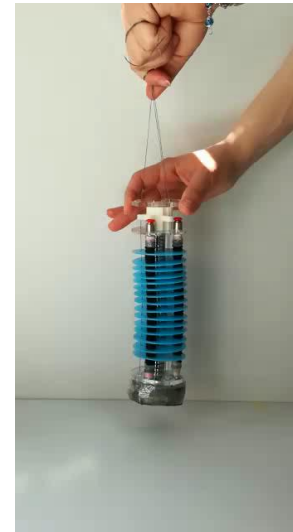
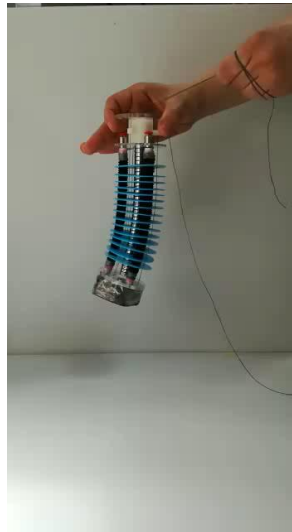
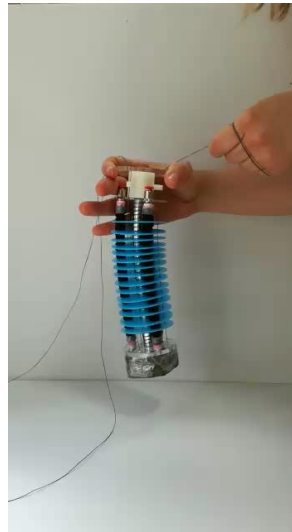
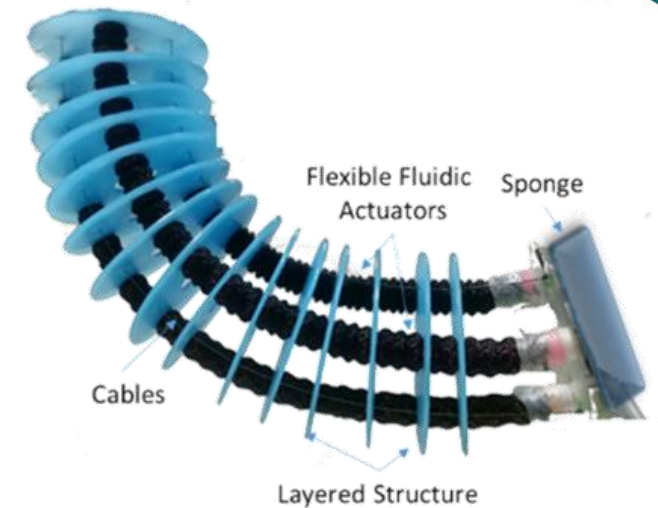
MODULAR APPROACH



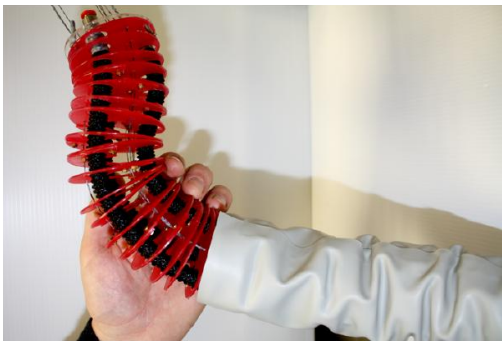
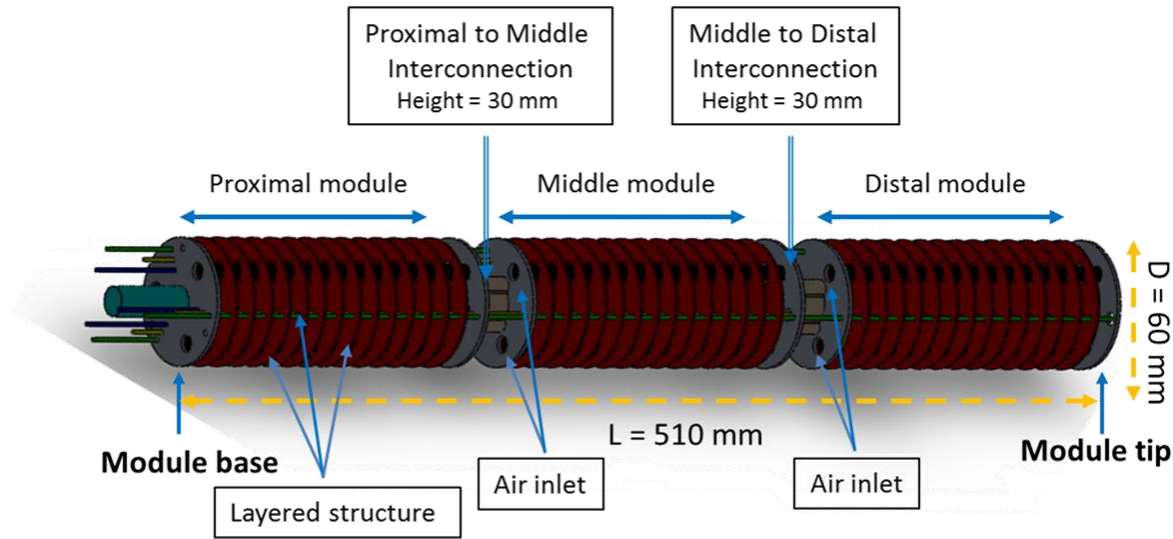
1. Bending on multiple planes

2. Contraction

3. Stiffness Modulation



I-Support Robot Arm



MODULAR APPROACH



CHANGABLE END-EFFECTOR



WATER-PROOF COVER

Pilot Studies: 2 Phases in 2 Sites



Pilot site 1 – Fondazione Santa Lucia (Rome, Italy)

Primary users: I-Support target user group

Secondary users: care staff, healthcare professionals

Pilot site 2 – Bethanien Hospital (Heidelberg, Germany)

Primary Users: I-Support target user group

Research team members for heuristic evaluation of usability

First Pilot Study: Results for Secondary Users

Summary: The scores at TAM 8 dimensions suggest a **positive experience**.

Secondary users seem rather convinced of the **ease of use**, **output quality**, and **job relevance** of the I-Support system

Scores of “intention to use” seems promising, as it suggests the actual **feasibility** of using the system in a **real healthcare context**

Technology Acceptance Model (TAM)

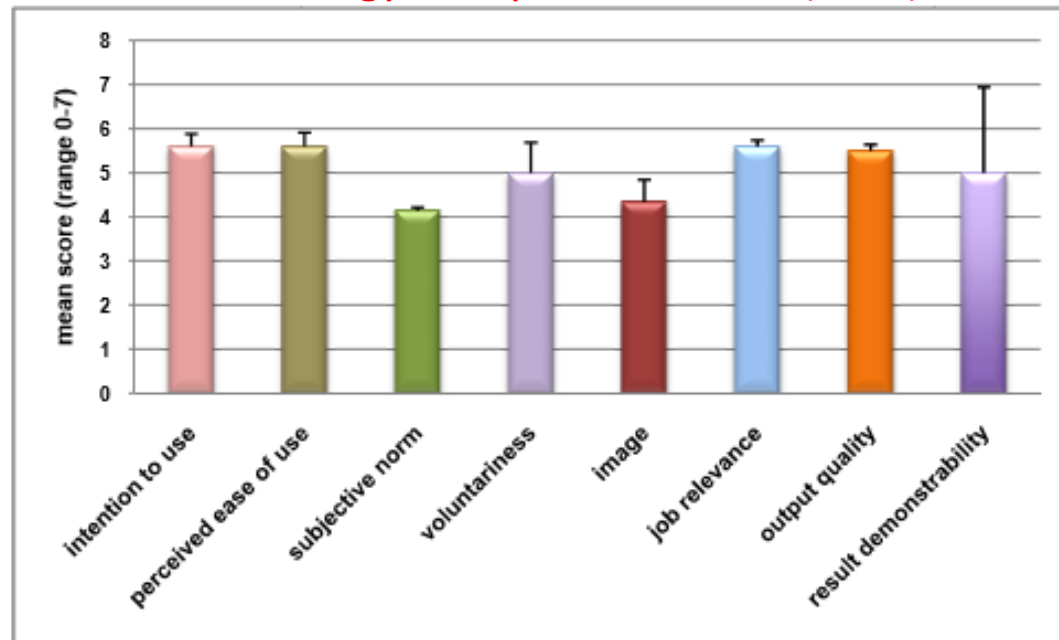


Figure 7: secondary users' feedbacks as expressed by TAM eight dimensions

Touch & Feel custom made questionnaire

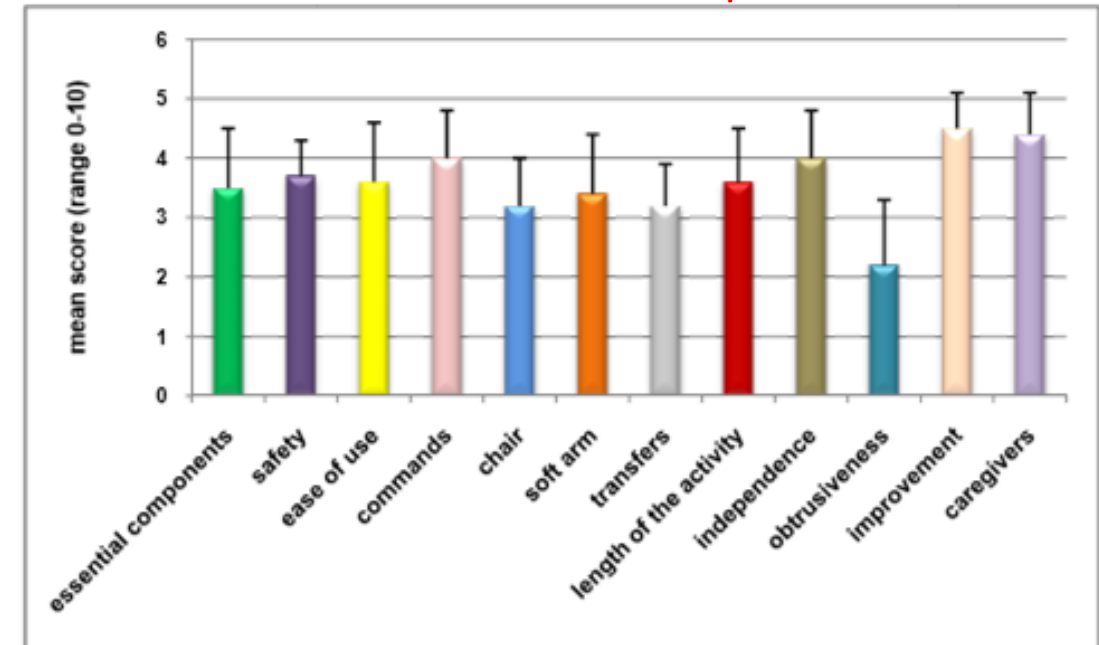


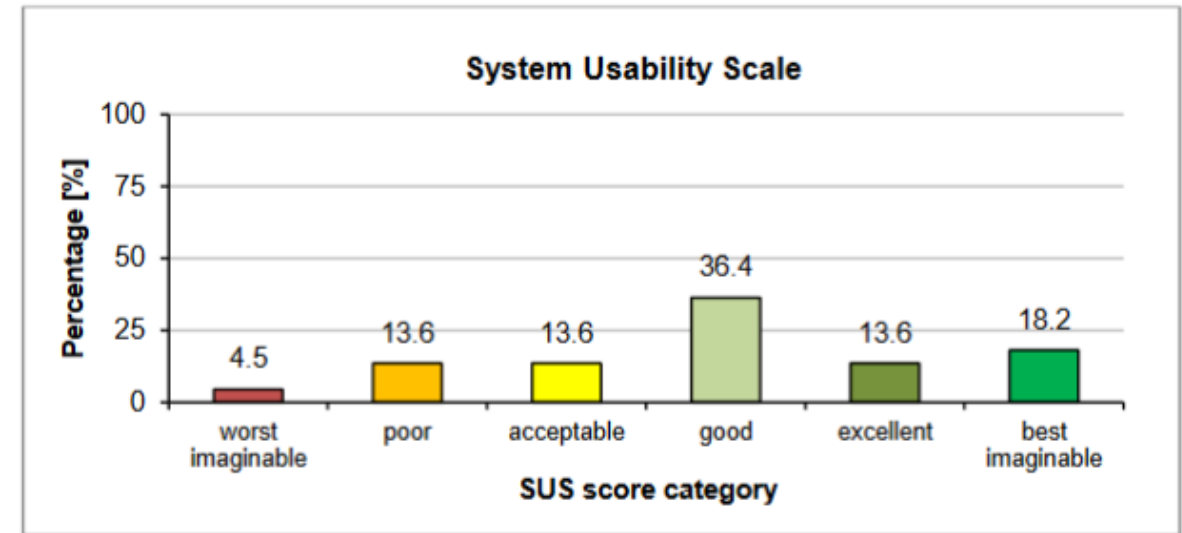
Figure 8: secondary users' feedbacks as expressed by the Touch & Feel questionnaire

Note: as a reverse item, “obtrusiveness” is coded in the opposite direction, thus it must be reverse coded (i.e., reported score of 2.2 is equivalent to 2.8)

Second Pilot Study: Results from primary users (@ Bethanien)

SCENARIO: Water pouring on the back region

- Within- and between-subject study
- 3 different operation modes:
 - autonomous operation
 - shared control
 - tele-manipulation
- Evaluation of:
 - effectiveness in showering the upper back region
 - user satisfaction in the study sample of primary users with and without cognitive impairment



Summary:

Autonomous operation seemed to be not only the **most effective**, but also the **most preferred** operation mode in the primary users of the I-SUPPORT system.

A substantially **lower effectiveness with user-controlled operation** modes has been observed, which might, however, be increased with a prolonged use or a more extensive training prior to usage.

Second Pilot Study: Results from Primary Users (@ Santa Lucia)

SCENARIO: Water pouring on the leg region

- Within- and between-subject study
- 3 different operation modes:
 - autonomous operation
 - shared control
 - tele-manipulation
- Evaluation of:
 - effectiveness in showering the leg region
 - user satisfaction in the study sample of primary users with and without cognitive impairment

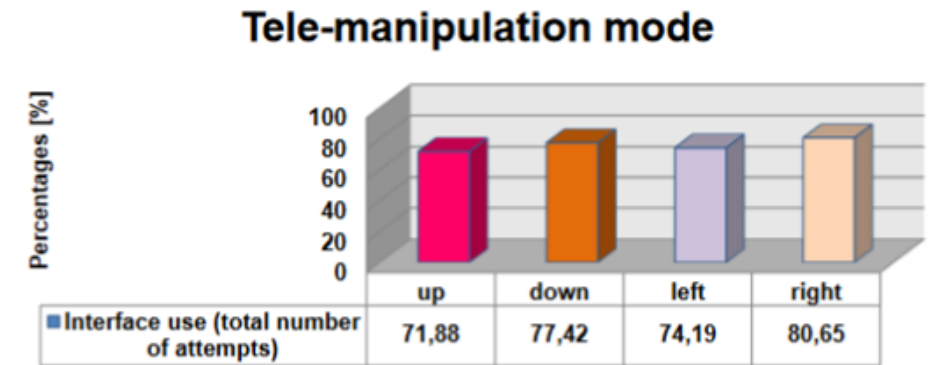


Figure 7: correct use of the tablet interface rates for tele-manipulation mode

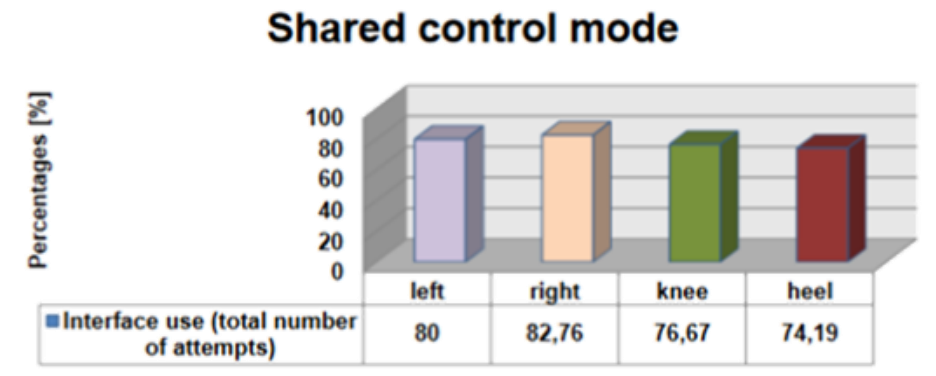


Figure 6: correct use of the tablet interface rates for shared control mode

Summary:

In both **Shared control** and **Tele-manipulation** mode, the rate of correctness of the user-interface interaction was not optimal, but promising.

Shared control: range of 74%-83% (mean percentage 78.4%),

Tele-manipulation: range of 72%-81% (mean percentage 76.03%).

Some issues related to the user-interface interaction through touchscreen

Second Pilot Study: Results from Secondary Users

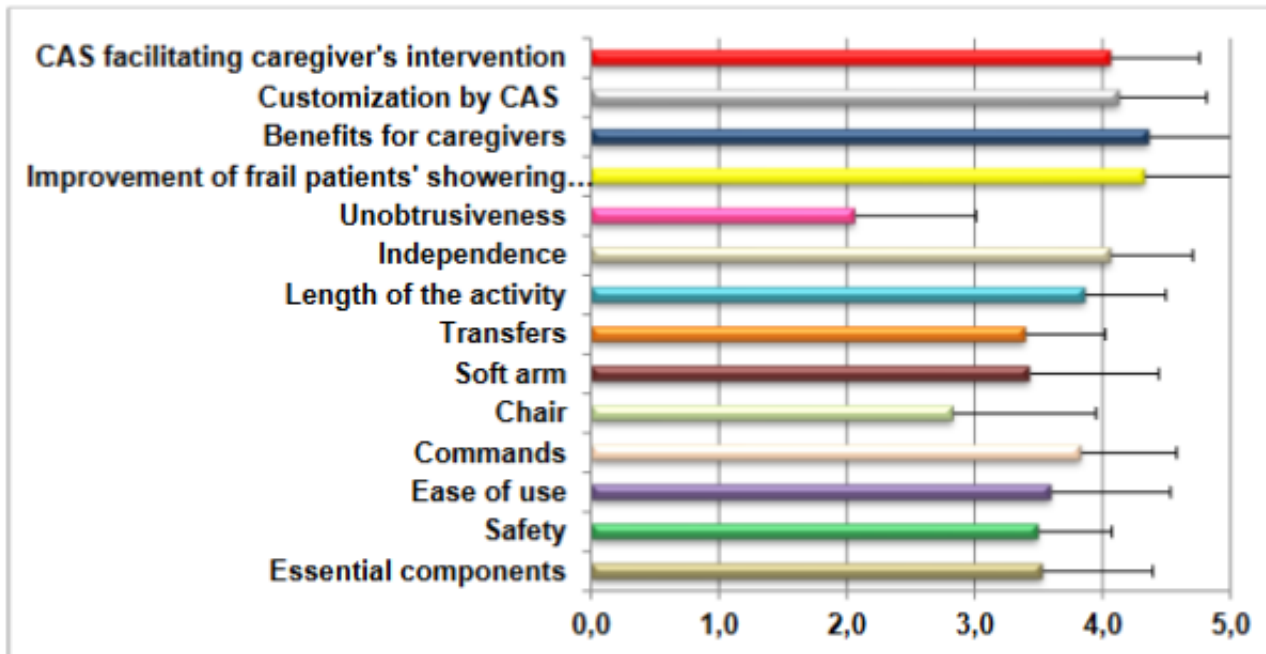


Figure 10: Secondary users' feedbacks as expressed by the Touch & Feel questionnaire

Note: as a reverse item, "unobtrusiveness" is coded in the opposite direction, thus it must be reverse coded (i.e., the reported score of 2.1 is equivalent to 2.9)

"[...] I think it could help elderly people taking shower autonomously and safely"

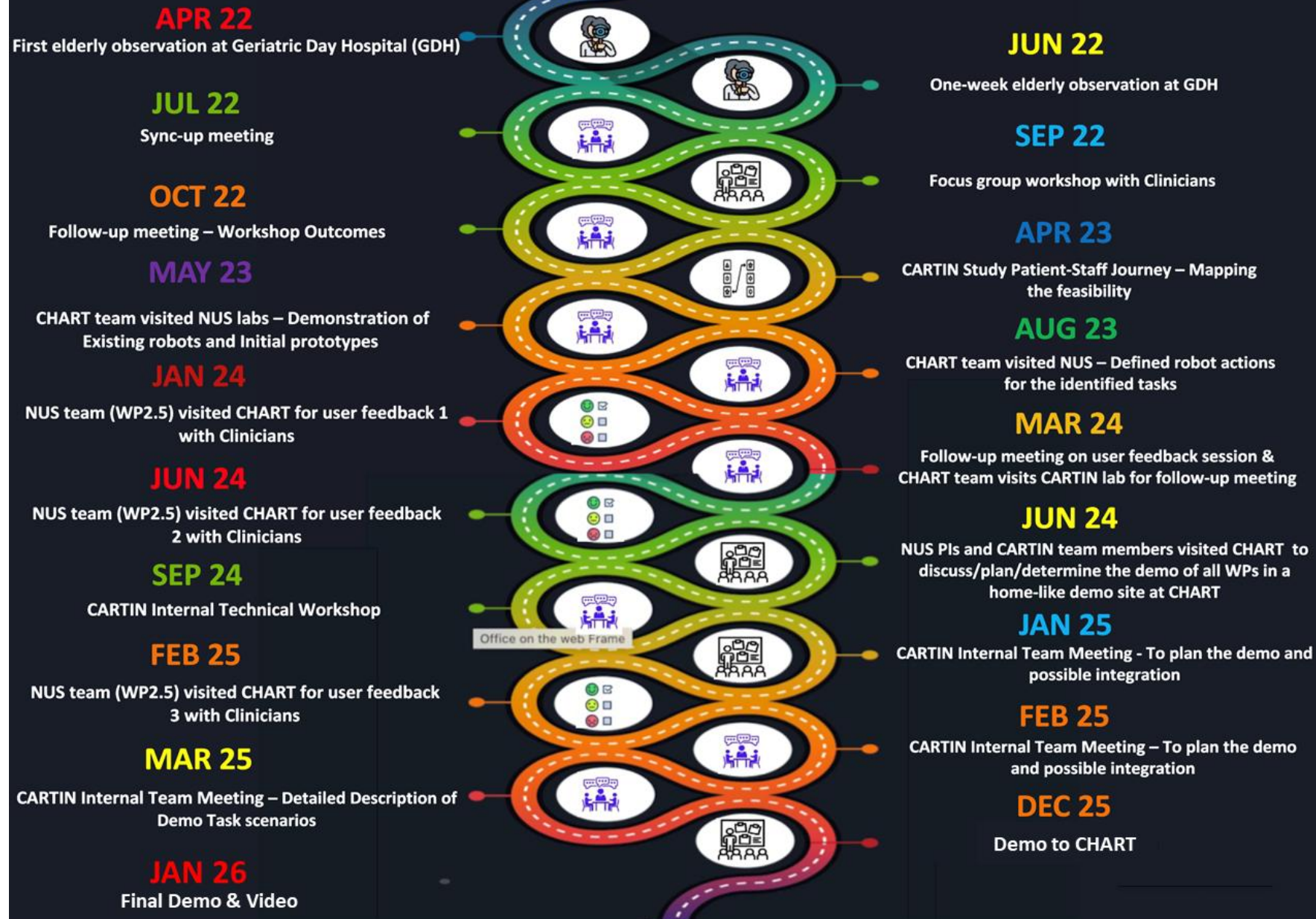
Senior Primary User

Example of Application of User-centred Design to an Assistive Robot for Eldercare

CARTIN

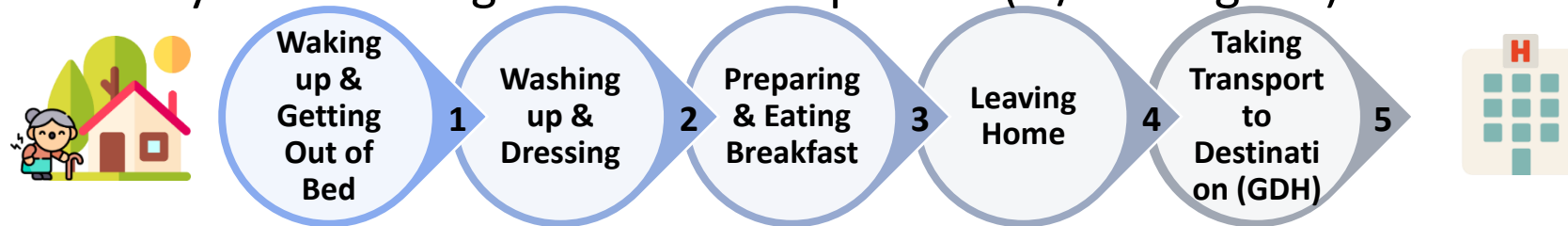


UCD (USER-CENTRED DESIGN) - COLLABORATION WITH CHART - ROADMAP



Key User Scenarios, for Patients and Caregivers

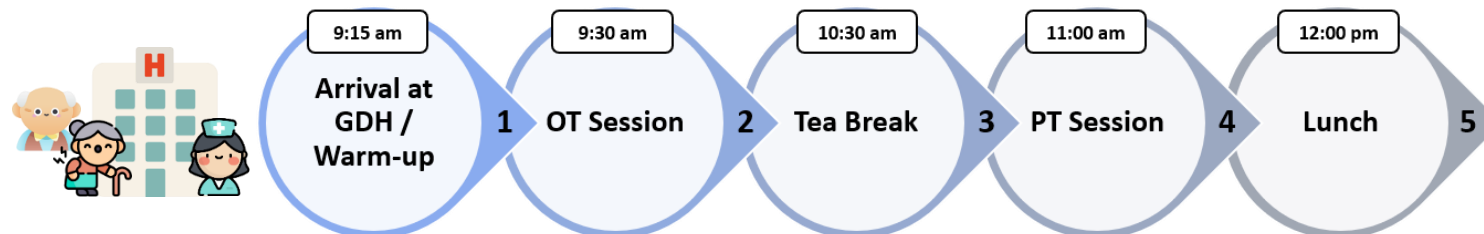
Patient's Journey – A morning in the life of a patient (w/o caregiver)



Mdm X, a 73-year old, has recently undergone a hip replacement surgery. She has been diagnosed with early dementia and goes to the Geriatric Day Hospital (GDH) once a week from 9am to 5pm for therapy to walk and to learn how to live independently as she lives at home alone.

Mobility: Some assistance required (uses mobility aids), **Cognitive:** Early stage of dementia, **Physical:** Suffers from mild hand tremors/Parkinsons, has poor strength/balance issues, **Language:** Only speaks Mother Tongue, limited understanding of English, **Environment:** Residential (alone)

Staff/Caregiver Journey – Assisting Patient at GDH



Mr Y, a 75 year old, is the caregiver for Mrs Y who is recovering from a stroke and is largely chair/bed bound. She requires help in transferring, dressing and bathing. Mr Y experiences some strain in his back due to the physically intensive tasks required to care for Mrs Y. When Mrs Y visits GDH for her therapy, Nurse A would help to assist Mrs Y, together with Mr Y.

Mobility: No issues, **Cognitive:** No issues, **Physical:** Experiences back pains due to physically intensive tasks required during caregiving, **Language:** Speaks both Mother Tongue and English, **Environment:** Residential (with caregiver)

Assistance to Transferring the Patient

- Focus on sit-to-stand, stand-to-sit and assisted walking for elderly support

Sit-to-Stand



Stand-to-Sit



Assisted Walking



Assistive Soft Robot

Flexible Yet Load-Bearing Arm



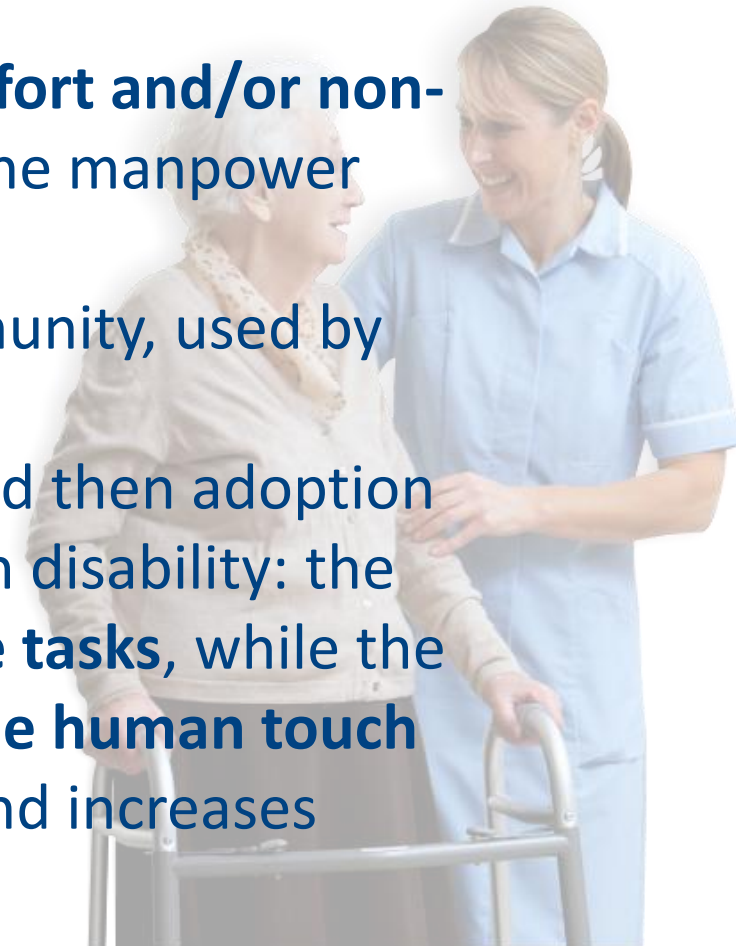
Motion capabilities



- **High vertical load capacity** (metallic skeleton mechanism)
- **Horizontal compliance** with bending and extension capability

Robotics as Assistive Technology

- Many manual back-breaking tasks that require **physical effort and/or non-ergonomic postures and movements** are taking a toll to the manpower situation in healthcare
- Robots for personal assistance, either at home or in community, used by caregivers
- User-centred design approach to increase acceptability and then adoption
- Proper **allocation of roles** of caregivers and individual with disability: the robot provides **physical support and relief from repetitive tasks**, while the caregiver provides **supervision, expertise and maintain the human touch**
- Proper **human-robot role allocation** gives value to both and increases adoption opportunities for robotic assistive technologies



As a robotic engineer, what do you think most challenging in designing an assistive robot?



Nobody has responded yet.

Hang tight! Responses are coming in.

Summary

Questions ?

- Healthcare robotics include robots used as Assistive Technology
- Assistive Technology (AT) is part of a rehabilitation process and supports individuals with disability in ADLs (Activity of Daily Life)
- AT requires design methods that include acceptability and involve users – the same for robotic assistive technologies
- Assistive robotic technologies include robotic manipulators to use as fixed workstations or wheelchair-mounted, smart wheelchairs, mobile robots, exoskeletons, robotic prostheses for upper and lower limb